

g0-sense — Design Document

ai-ee v1 pipeline

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1 Board and Run Metadata

Field	Value
board	g0-sense
workspace	boards/g0-sense
phase	P8
gate status	all 4 recorded gates pass
state created	2026-08-27T06:35:38
state updated	2026-08-27T19:08:37

2 Overview

Brief: g0-sense (unattended container run, 2026-08-27)

A small 2-layer USB-C powered temperature/humidity sensor node - a product-scope board (no mode token: design normally), meant to be sent to JLCPCB as-is.

- Power: USB-C receptacle used for POWER ONLY (5 V VBUS; 5.1 k CC pull-downs so any USB-C source or a legacy A-to-C cable enables VBUS; D+/D- unconnected). Sensible input protection for a USB-powered device (VBUS TVS, resettable fuse or equivalent, reverse/inrush as the architect sees fit).
- 3.3 V rail from a fixed LDO, ≥ 300 mA, JLC Basic/Preferred part if one fits (AMS1117-3.3 class is fine), with proper input/output capacitors.
- MCU: STM32G030F6P6 (TSSOP-20) on the internal oscillator - no crystal. Decoupling per the datasheet, NRST with a push button, BOOT0 handled.

- Sensor: Sensirion SHT40 (or the SHT4x family member in JLC stock) on I2C with pull-ups; give it a thermal isolation slot/cutout or at least keep it away from the LDO and MCU heat.
- Interfaces: 4-pin SWD header (0.1 in), 4-pin UART header (0.1 in: GND, 3V3, TX, RX) for readout/logging, one Qwiic/STEMMA-QT connector (JST SH 1.0 mm 4-pin, GND/3V3/SDA/SCL) sharing the sensor I2C bus.
- One user LED (GPIO, series resistor) and one power LED.
- 2 layers, 1.6 mm FR-4, HASL, green; JLC economy PCBA for the SMD parts, prototype qty 5. Prefer JLC Basic parts; Extended parts only where the brief names the part (MCU, sensor, connectors).
- Size: whatever an honest layout wants, roughly 35 x 25 mm or smaller; four M2 mounting holes if they fit without hurting the layout.
- No battery, no mains, no USB data, no RF.

Stackup

Chosen stackup: JLC2313_1.6 (reference/stackups.yaml, available: true)

3 Requirements

requirements - g0-sense

P0 requirements distilled from `brief/brief.md` (2026-08-27, unattended container run).

1. Function

Mode declaration: the brief carries NO mode token, so mode = none - design normally. Scope = product-scope per the brief's own words ("a product-scope board ... meant to be sent to JLCPCB as-is"): protection, filtering, connectors, thermal and enclosure-fit are all in scope, and their absence is a reviewer finding. Binding: the stated ~35 x 25 mm size is a SOFT preference, not a hard cap (see section 5). Stage under study: none.

The board is a small 2-layer USB-C powered temperature/humidity sensor node. A USB-C receptacle supplies 5 V (power only, no data); a fixed 3.3 V LDO powers an STM32G030F6P6 MCU running on its internal oscillator, which reads a Sensirion SHT4x sensor over I2C. The same I2C bus is exposed on a Qwiic/STEMMA-QT connector; readout/logging is over a 4-pin UART header; programming/debug is over a 4-pin SWD header. One user LED and one power LED. Prototype quantity 5, JLC economy PCBA, meant to be ordered as-is.

2. Interfaces

- USB-C receptacle, POWER ONLY: 5 V VBUS input; 5.1 k pull-downs on CC1 and CC2 (so any USB-C source or a legacy A-to-C cable enables VBUS); D+/D- unconnected; no USB data.
- SWD header: 4-pin, 0.1 in pitch. ASSUMED: pins GND, 3V3, SWDIO, SWCLK (exact order per architect; brief states only "4-pin SWD header (0.1 in)").
- UART header: 4-pin, 0.1 in pitch: GND, 3V3, TX, RX - for readout/logging. ASSUMED: TX/RX are named from the MCU's perspective (TX = MCU transmit).
- Qwiic/STEMMA-QT connector: JST SH 1.0 mm 4-pin, GND/3V3/SDA/SCL, sharing the sensor I2C bus. I2C pull-ups required on the board (values per architect).

- User LED: one, driven from a GPIO through a series resistor.
- Power LED: one (which rail it indicates - VBUS or 3V3 - is per architect; the brief does not say).
- NRST push button (momentary reset).
- BOOT0: handled (method - strap, option bits, or jumper - per architect; the brief requires only that it is handled).
- Four M2 mounting holes, CONDITIONAL: only if they fit without hurting the layout (see section 5).

3. Power

- Input: USB-C VBUS, 5 V nominal, device draws < 1 A total. No battery, no charging, no mains.
- VBUS input protection required (product scope, brief-stated): TVS, resettable fuse or equivalent; reverse/inrush handling as the architect sees fit - the brief delegates the exact topology.
- 3.3 V rail: fixed LDO, ≥ 300 mA. JLC Basic/Preferred part preferred; AMS1117-3.3 class acceptable. Proper input/output capacitors per the chosen part's datasheet.
- Rail budget (GUESSES - to be replaced by the architect's numbers):
 - STM32G030F6P6 on internal oscillator: ~ 5 -15 mA (guess)
 - SHT4x: < 1 mA average, single-digit mA peaks during measurement (guess)
 - LEDs: ~ 2 -5 mA total (guess)
 - Qwiic downstream devices: not budgeted by the brief - see open question 2 (recommended default: reserve 100 mA).
 - On-board total is tens of mA; the brief's ≥ 300 mA LDO floor leaves the headroom for Qwiic loads.

4. Environment

Not stated in the brief. ASSUMED: indoor, room-ambient use (roughly 0-40 C), bare board, no ingress or vibration requirements (see open question 1). The one stated thermal/environmental requirement is layout-level: the SHT4x gets a thermal isolation slot/cutout, or at minimum is kept away from LDO and MCU heat, so self-heating does not corrupt the measurement.

5. Size & mounting

- Outline: "whatever an honest layout wants, roughly 35 x 25 mm or smaller". This is a SOFT preference - no HARD cap. It must NOT bind at P5 board_init; the layout earns the final outline, and ~ 35 x 25 mm is a target to aim for, not a box to squeeze into (see open question 5).
- Mounting: four M2 mounting holes, CONDITIONAL - "if they fit without hurting the layout". The brief itself allows dropping or reducing them in favor of layout honesty; record that decision if taken.
- Height: no limit stated.

6. Quantity & budget

- Quantity: 5 boards, prototype run, assembled.
- Unit cost: no target stated (see open question 3). Cost posture from the brief: JLC economy PCBA; prefer JLC Basic parts; Extended parts only where the brief names the part (MCU, sensor, connectors).

7. Assembly

- JLC economy PCBA for the SMD parts.
- ASSUMED: single-sided SMD assembly (economy PCBA norm, and this part count does not need the second side).
- Through-hole parts: economy PCBA assembles SMD only, so the two 0.1 in headers are ASSUMED to ship unpopulated for owner hand-soldering (see open question 4).
- ASSUMED: NRST push button and the USB-C / JST SH connectors are SMD, JLC-assemblable variants where stock allows, so they ride the PCBA run.
- Board: 2 layers, 1.6 mm FR-4, HASL, green (brief-stated).

8. Compliance/safety flags

Each flag checked against the brief; none applies:

- Mains voltage: NOT APPLICABLE - brief states "no mains"; input is USB 5 V.
- Battery: NOT APPLICABLE - brief states "no battery"; no charging circuitry anywhere in scope.
- Motors: NOT APPLICABLE - no motor or actuator in the brief's scope.
- >30 V: NOT APPLICABLE - the highest voltage on the board is USB VBUS at 5 V nominal.
- High current (>3 A): NOT APPLICABLE - the brief describes a USB 5 V, < 1 A device.
- RF transmit: NOT APPLICABLE - brief states "no RF"; no radio, and no USB data either ("D+/D- unconnected").

9. Open questions

Closed-form, each with a recommended default. Unattended run: the orchestrator answers these on the owner's behalf and records the answers.

1. Operating environment: is indoor, room-ambient use (about 0-40 C, bare board, no ingress/vibration requirements) correct? Recommended default: YES. 2. Qwiic 3.3 V budget: how much current may external Qwiic/STEMMA-QT devices draw from the 3.3 V rail? Recommended default: reserve 100 mA (fits under the ≥ 300 mA LDO floor together with the on-board load). 3. Unit cost: is there a hard per-board cost cap? Recommended default: NO hard cap - minimize via economy PCBA and Basic parts as the brief already directs. 4. Header population: should the two 0.1 in headers (SWD, UART) ship unpopulated for hand-soldering, given economy PCBA assembles SMD only? Recommended default: YES - footprints and holes present, headers not populated at assembly. 5. Size confirmation: is $\sim 35 \times 25$ mm truly a soft preference (the layout may exceed it if honesty demands), not an enclosure-driven hard cap? Recommended default: SOFT preference, per the brief's own wording ("whatever an honest layout wants").

4 Architecture

Decisions of record (state.json)

Phase	What	Why
P0	unattended default: environment = indoor bench/room ambient, 0-40 C operating, bare board, no ingress or vibration rating	Q1 of requirements section 9; brief names no environment. Most conservative common-practice choice for a USB-powered indoor sensor node; parts will still be picked at industrial -40..85 C grade where that is the stocked option, so the assumption cannot make the board wrong.
P0	unattended default: Qwiic/STEMMA-QT down-stream budget = 100 mA reserved on the 3V3 rail	Q2; brief does not budget it. 100 mA covers typical Qwiic sensor/display loads and still fits inside the brief's ≥ 300 mA LDO floor with margin for the on-board tens of mA, so it sizes the LDO and its thermals honestly without inflating the part.
P0	unattended default: no hard per-board cost cap; minimize cost via JLC economy PCBA and Basic parts as the brief already directs	Q3; brief states no budget. Cost is already bounded by the brief's own Basic-parts-preferred rule; inventing a cap could force a worse engineering choice.
P0	unattended default: the two 0.1 in headers (SWD, UART) ship UN-POPULATED - plated through-holes only, hand-solderable; only SMD parts go to JLC economy PCBA	Q4; brief says 'JLC economy PCBA for the SMD parts'. JLC economy assembly is SMT-only, so THT headers are BOM 'do not populate / customer supplied'. Standard practice for dev/sensor boards and it keeps the assembly quote in economy tier.
P0	unattended default: $\sim 35 \times 25$ mm is a SOFT preference, not an enclosure-driven hard cap; no HARD outline cap binds at P5	Q5; the brief says 'whatever an honest layout wants, roughly 35×25 mm or smaller' and names no enclosure. Confirms requirements.md section 5; placement must not optimize to fit an unearned number (the bb-buck failure mode).
P1	unattended default: LDO = AMS1117-3.3 class in SOT-223 (JLC Basic, C6186); governing thermal design point is the brief's literal ≥ 300 mA rated case (0.51 W), not the ~ 150 mA realistic load	research/ldo.md escalated the SOT-223-vs-SOT-23-5 tradeoff; research/power.md resolved it with numbers: AMS1117 T_j 71-86 C at 40 C ambient and ≤ 115 C even in the 500 mA abuse case, while AP2112K-class SOT-23-5 (θ_{JA} 184 C/W) hits T_j 134 C at the rated case with no copper fix available. The Qwiic port is an unfenced 3V3 output, so the rated case is reachable in the field. Basic part, 1.1M stock, \$0.20.
P1	unattended default: a sub-6 V TVS clamping voltage is NOT a requirement; SMAJ5.0A/SMF5.0A-class (standoff 5.0-5.5 V, $V_c \sim 9.2$ V at I_{pp}) is accepted	research/protection.md flagged that no passive TVS clamps under 6 V. Nothing on VBUS is 6 V-rated: the only VBUS consumers are the LDO (AMS1117 V_{in} abs max 15 V), the input caps (16 V+ parts) and the PTC (16 V). Everything at 3.3 V sits behind the LDO. A ~ 9.2 V clamp during a surge is therefore inside every VBUS part's rating - the original < 6 V framing was an over-tight derived spec, not a real one.

Phase	What	Why
P1	unattended default: NO series reverse-polarity element (no Schottky, no P-FET ideal diode) on VBUS	Two independent reasons, both recorded in research: (a) a compliant USB-C cable cannot present reverse polarity (protection.md), and (b) research/power.md computed the end-to-end dropout budget and found a series Schottky's 125-190 mV would consume the entire AMS1117 headroom at the 4.75 V source corner (+0.11 V worst-corner margin as designed). The unidirectional TVS already crowbars a reverse fault. Adding it would make the board worse, not safer.
P1	unattended default: USB-C receptacle = 16-pin power-only part whose THT shield/mounting legs are accepted; JLC assembles it in the Economic PCBA tier	research/connectors.md correctly flagged that every 16P part in this class has THT shield legs and asked whether economy PCBA can place them. Verified directly on the JLCPCB part page for C165948 (TYPE-C-31-M-12): Assembly Process = SMT, Mount Type = Surface Mount Right Angle, and the part is listed as supported by BOTH Economic and Standard assembly. So the legs are not an assembly blocker. Part is JLC Extended - permitted, the brief names the connector.
P1	unattended default: SHT4x on-chip heater (75 mA peak) IS budgeted in the rail tree at <=10% duty	research/power.md's call, accepted: including it costs nothing (peak +3V3 still 185 mA, 240 mA with 30% headroom, under the 300 mA LDO floor and at 39% of the 500 mA CC-blind USB entitlement), while omitting it would turn a documented firmware feature into a hardware ECO later.
P1	unattended default: NRST tactile button - prefer the JLC Basic 5.1x5.1 mm part over the brief's smaller 3x2.5/4.2x3.2 mm size class if the layout has room; size class is a preference, Basic status and stock depth are the tiebreak	research/connectors.md Q2: the only Basic button is 5.1x5.1 mm (C318884, 962k stock, \$0.02); the closest-size alternative carries a documented lib_pull symbol gotcha in this repo's own LEARNINGS. The brief never fixed the button size, and the board size is soft, so paying ~2.9 mm of edge for a Basic, gotcha-free part is the conservative trade. Architect may overrule at P2 if placement suffers.
P1	unattended default: LED colours - user LED green, power LED red; both 0603, series resistors sized from the measured Vf band per colour	research/connectors.md Q3 left the assignment open. Red is the conventional power indicator and is the JLC Basic option with the low Vf band (1.8-2.4 V), which gives the largest resistor and the lowest current draw for an always-on LED. Green for the user LED is the conventional 'activity' colour; it is Extended in the KT-0603 family, which is acceptable at one extra unique part, and its 2.6-3.1 V Vf is fine from 3.3 V at a few mA.
P2	research task block-mcu-1 closed over 1 unverified draft record(s): mcu-boot-strap-only-when-option-bytes-consult-pin	-accept-drafts: the slot ships without this knowledge

Phase	What	Why
P2	unattended default: BOOT0 gets a populated 10k pull-down on PA14 (pin 19, shared with SWCLK) - the board does NOT rely on the unverified factory-option-byte claim	The research record mcu-boot-strap-only-when-option-bytes-consult-pin was REFUTED by a fresh second reader: its load-bearing claim (factory 0xDFFFE1AA / nBOOT_SEL=1 makes the BOOT0 pin ignored) rests solely on a community.st.com forum page, and RM0454/AN2606 could not be acquired - st.com times out from this container (verified twice by curl, once by an agent). Rather than design on unverified knowledge, the board is made correct under EITHER option-byte state: a 10k pull-down guarantees BOOT0=0 at reset so the part boots main flash even if nBOOT_SEL=0. Cost is one 0402 resistor; 330 uA load on SWCLK is negligible to any debugger's push-pull driver, and the datasheet-confirmed internal pull-down on PA14 in debug mode already pulls the same way. This closes the gap without the source.
P2	unattended default: I2C pull-ups are 1.5 kOhm (not the 2.2 kOhm the P1 fragment proposed)	The draft record sht4x-i2c-pullup-bus-cap was refuted for recommending 2.2k over its own declared 150-200 pF bus budget: UM10204 Eq 1 at tr=300 ns gives $R_p(\max) = 354/C_b[\text{pF}]$ kOhm, i.e. 1.77k at 200 pF, so 2.2k needs $C_b \leq 161$ pF and misses the Fast-mode rise time on a cabled Qwiic bus. 1.5k holds the ceiling to 236 pF and clears both floors (967 Ohm from UM10204 Table 10 at 3.3 V/3 mA, 390 Ohm from the SHT4x datasheet). Value independently recomputed by a second reader. architecture/constraints.json and the P4 schematic must use 1.5k.
P3	unattended default: AMS1117 SOT-223 tab is treated as VOUT (+3V3) - the thermal pour is a +3V3 pour, not an isolated island	The datasheet-extractor honestly refused to assert it: the 'TAB IS OUTPUT' label on page 1 sits against the TO-252 view, and no text restates it for SOT-223. Resolved as tab=VOUT because (a) the datasheet's own 15 C/W junction-to-TAB figure means the tab is the die-attach pad, which on this family's leadframe is the VOUT lead, (b) the KiCad standard symbol/footprint pair the librarian pulls ties tab to VOUT, so schematic and layout agree by construction, and (c) it is the safe direction either way: if the tab were floating, a +3V3 pour on it is still harmless, whereas assuming 'floating' and isolating the copper would throw away the thermal design the 0.51 W design point depends on. Flagged for the schematic reviewer to confirm against the pulled symbol.

Phase	What	Why
P3	unattended default: the SHT4x 20 V/ms VDD slew limit needs NO extra bulk capacitance - carried item CLOSED, not escalated	Two independent second readers read the SHT4x datasheet Table 4 slew row and quoted its condition verbatim: 'Voltage changes on the supply between VDD,min and VDD,max. Faster slew rates may lead to a reset' (min cell empty). That bounds supply changes DURING operation, not the cold-start ramp from 0 V - and a reset during power-on is the intended behaviour anyway (the part has a POR). In operation the 22 uF tantalum on +3V3 holds any load-step slew orders of magnitude below 20 V/ms. Adding bulk to slow a cold start would also fight the USB-C ≤ 10 uF attach limit upstream. No hardware change; note it for bring-up.
P4	unattended default: WAIVE schematic-review W1 (no ESD array on the J2 Qwiic SDA/SCL) - ship the board without a TVS/ESD array on the I2C bus	Product-scope connector checklist allows met-or-waived. Waived on four grounds: (a) the Qwiic/STEMMA-QT ecosystem norm (Spark-Fun/Adafruit reference boards) is a bare 4-pin JST SH with no array, so a user-facing part behaves as the ecosystem's users expect; (b) STM32G030 PA11/PA12 and the SHT40 both carry internal ESD structures and are 2 kV HBM rated, which is the standard handling rating for an indoor bench device; (c) the recorded environment is indoor bench/room ambient with no ingress rating, not a field-exposed port; (d) adding a 4-channel array now means a new Extended MPN plus setup fee, a fresh library pull and footprint verify, and re-opening a green schematic at P4 - real regression risk against a warning-severity, field-reliability-only exposure that cannot prevent bring-up. Recorded in reports/erc-waivers.md and shown at H2.

Phase	What	Why
P4	unattended default: CLOSE the carried verify item 'SHT4x VDD slew ≤ 20 V/ms vs AMS1117 startup ramp' (schematic-review W2) as accepted risk - no hardware change	Three findings close it. (1) PROVENANCE IS ON FILE, contrary to the reviewer's note: the 20 V/ms figure is research/records/sht4x-vdd-slew-power-up.yaml, sourced to research/sources/HT_DS_Datasheet_SHT4x.pdf p9 Table 4 and read visually TWICE (researcher + fresh second reader, maturity verified). The reviewer grepped parts/C2909890.json (the extraction JSON) and the LCSC PDF, which is a different artifact. (2) THE WORST-CASE NUMBER IS AN UPPER BOUND THAT CANNOT OCCUR: I_{limit}/C (~ 1 A into ~ 27 uF, 40-68 V/ms) assumes the LDO sources its full current limit into the output cap independently of its input. The AMS1117 is a series pass follower - during a cold start VOUT cannot rise faster than VIN minus dropout, so +3V3's edge is set by the USB source's VBUS ramp through C2 = 10 uF, not by the LDO's current limit. (3) THE NAMED FAILURE MODE IS BENIGN AT COLD START: the datasheet says only 'faster slew rates MAY lead to a reset'; the second reader explicitly flagged 'or hung until re-powered' as an unsupported clause the researcher added. A reset during the power-up ramp lands the part in exactly the state a power-up is supposed to produce - there is no state to lose at $t=0$. The limit's real bite is in-operation supply excursions and hot-plugging a DOWNSTREAM breakout onto a live Qwiic rail, which is the breakout's decoupling problem, not this board's. HARDWARE ALTERNATIVES REJECTED: a series R at U3 VDD to slow the local ramp is disqualified by the SHT4x on-chip heater (75 mA peak would drop 7.5 V across the 100 ohm needed for a 100 us tau); raising C3 above ~ 50 uF to force I_{limit}/C under 20 V/ms buys nothing against (2) and costs a part change on a green schematic. BELT AND BRACES (documentation, not copper): the firmware note 'issue an SHT4x soft reset (0x94) at init, per Sensirion's own recommended start-up sequence' goes into the design doc and fab/README, and the Qwiic hot-plug caveat is recorded as a known limit.

Phase	What	Why
P4	unattended default: carry a P9 REQUIRED check - CPL rotation must be validated for every polarized 2-pin part (D1 TVS SOD-123, D2 power LED, D10 user LED, C3 tantalum CASE-A), not only for J1	schematic-review OPEN item. decisions.md #13 carried a CPL rotation check for J1 alone; the KiCad<->JLC 0-degree-reference offset (reference/jlc_rotations.csv, LEARNINGS/S8) bites every polarized package, and a reversed tantalum on 3V3 or a reversed TVS on VBUS is a dead board, not a cosmetic error. dfm_check.py owns polarity/rotation validation - at P9 its findings for D1/D2/D10/C3 must be read explicitly and each confirmed against the footprint pin-1 marking, not just accepted as a count.
P5	unattended default: P5 runs board_init -mounting-holes 4 -mounting-hole-fp MountingHole:MountingHole_2.2mm_M2 -margin 6 -outline auto; the four holes are explicitly RELEASABLE at P6 - the placement agent may move them with place.edit to the earned corners before board_edit -outline fit	Three linked calls. (1) HOLE SIZE: board_init's mounting-hole footprint was hard-coded to MountingHole.3.2mm_M3, but the brief and constraints ask for M2 (2.2 mm drill). I added an additive -mounting-hole-fp flag to board_init.py, default unchanged, so the M2 hole is stated rather than inherited (MountingHole_2.2mm_M2: NPTH 2.2 mm drill, 4.9 mm courtyard, which also serves the constraints' 4.0-4.5 mm annulus/head clearance). (2) MARGIN 6: -mounting-holes places holes at margin/2, so margin 6 puts the M2 centres 3 mm inside the edge - the standard M2 corner inset, and 1.9 mm of board between hole wall and edge. A smaller margin would push the hole courtyard across the board edge. (3) HOLES RELEASABLE AT P6: place.seed/place.anneal treat board_only footprints as immovable obstacles, so holes parked at the PROVISIONAL corners would fix the fitted bbox at the provisional size and defeat geometry-as-OUTPUT. They were auto-placed by a script at an arbitrary provisional corner, not by the owner, so the placement agent is authorised to move them with place.edit to the corners the earned layout wants - the agent-prompt rule against moving board_only refs exists to protect owner-placed mechanical fixtures, which these are not. Order follows the bb-mcu P6 recorded pipeline defect: place canonically, move the holes, board_edit -outline fit, and only THEN run the place gate (legality measures declared edges against the CURRENT outline, so gating before fit would spread the connectors to the provisional size).

Phase	What	Why
P5	unattended default: J1 USB-C footprint repaired at P5 - the four wide GND/VBUS pads restored to the vendor's own 0.60 mm land width, reopening the GND-VBUS gap from 0.100 mm to 0.200 mm	board_init's self-check DRC found two clearance ERRORS inside J1's own footprint: pad A1B12 [GND] vs A4B9 [VBUS] at 0.1002 mm and B1A12 vs B4A9 at 0.1000 mm, against the aiee_clearance_floor of 0.127 mm (JLC 2-layer 1 oz). J1 was never geometry-checked at P3 (fp_verify needs a datasheet-extract JSON and none exists for it). The librarian fetched the HRO Hroparts TYPE-C-31-M-12 drawing: its RECOMMEND P.C.B LAYOUT calls the four wide pads '4-0.60', i.e. 0.60 mm wide on a 0.8 mm pitch = a 0.20 mm gap, already above the floor. The pulled easyeda2kicad custom-pad polygons were the vendor's 0.60 mm, but carried a spurious 0.1 mm outline STROKE that inflated the effective copper to 0.70 mm and closed the gap to 0.100 mm. So this was a converter artifact, not a vendor deviation, and the fix RESTORES the manufacturer's land pattern rather than inventing a number: outer X vertices moved in by 0.05 mm per side on those four polygons only, pad centres / 1.30 mm length / the other 12 pads untouched. Contact tab is 0.20 mm wide (drawing callout 16-0.20+-0.05), so 0.60 mm still leaves 0.20 mm of fillet per side - the vendor's own allowance. Rationale recorded in the footprint descr, matching the P3 SHT40 die-pad precedent.

Phase	What	Why
P6	unattended default: P7 must hand-route the VBUS entry out of J1 - the 0.8 mm Pwr_0p8mm netclass cannot escape J1's 0.60 mm vendor pads on F.Cu at ANY placement, so route_critical owns it and Freerouting must not be asked to	The placement agent's route probe closed at 0.969 (62/64) and root-caused the two residuals: J1's VBUS pads are 0.60 mm wide with 0.20 mm gaps to their GND neighbours, so a 0.8 mm track (the VBUS netclass width rules_gen derived from the 1.5 A PTC-dwell case) cannot leave them on the top layer without breaking the 0.127 mm clearance floor - it is a geometric impossibility, not a placement defect, and no amount of re-placing fixes it. A via-in-pad escape to B.Cu was proven DRC-clean on a scratch copy; Freerouting cannot place via-in-pad, so this is route_critical's job before the autorouter runs. THREE THINGS P7 MUST WEIGH, in this order: (1) a short NECKED escape (fab-minimum width for ~1 mm out of the pad, widening to 0.8 mm once clear of the connector) is the ordinary way this is done and needs no via - but it violates the ERROR-severity aiee_pwr_width_VBUS rule, so taking it means a rule EXCEPTION with recorded evidence (the pad itself is 0.60 mm, so the 0.8 mm rule is already wider than the conductor the current must pass through), not a silent neck; (2) via-in-pad to B.Cu carves the B.Cu GND pour under the connector and JLC will wick solder into an unfilled 0.3 mm via on a mechanically loaded connector pad - if this route is taken it needs an order remark and a fab/README note; (3) the two VBUS pads are ganged, so each carries half the fault current. Do NOT resolve this by editing the netclass or the floor to make DRC quiet.

Phase	What	Why
P7	unattended default: J1 VBUS escape resolved as pour-carried copper (a measured third option), no rule exception and no via-in-pad	route_critical -pad-window measured J1 A4B9/B4A9 escape windows at 1.315 mm, comfortably above the 0.8 mm Pwr_0p8mm requirement, which retires the P6 premise that the escape was geometrically impossible - the impossibility applied to a 0.8 mm TRACK leaving the pad, not to the pad's available escape corridor. The F.Cu VBUS pours connect the pads with a solid zone connection and a ~1.45 mm fill band east of the pad column, so the conductor carrying the 1.5 A PTC-dwell fault current is WIDER than the 0.8 mm the rule asks for, not narrower. aiee.pwr_width_VBUS conditions on A.Type=='track', so zone fill is outside its scope by construction - the gate passes because no VBUS track anywhere is below 0.8 mm, not because a rule was relaxed. Neither P6 option was taken: no neck (so no ERROR-severity rule exception to record) and no via-in-pad on J1 (so no unfilled via in a mechanically loaded connector pad, and no JLC solder-wicking remark needed in fab/README). The two F.Cu VBUS pours are joined by a 0.8 mm B.Cu bridge on 2+2 vias placed in open pour with holes clear of every pad edge. VERIFIED BY THE ORCHESTRATOR, not taken on the router's word: plane_repair -flag-only on the final board reports VBUS as 1 electrical group (the two 8.4 and 7.8 mm² F.Cu components joined through the bridge), and kicad-cli DRC reports 0 violations / 0 unconnected / 0 parity.
P7	unattended default: the B.Cu VBUS bridge does NOT split the GND plane - verified on the final board, not inherited from the pre-bridge run	The 0.8 mm B.Cu VBUS bridge crosses the layer whose whole area is the GND pour, which is precisely the 'viasless pour-channel' blind spot the playbook lists as a router duty. The router's own plane_repair2 run was restricted to +3V3, so it never re-examined GND, and the only GND-clean evidence on disk (plane_repair.json: GND B.Cu 1 group, split False) predated both the snapshot restore and the bridge. I re-ran plane_repair.py -flag-only (never writes) on the final board: GND B.Cu = 1 group, split False, 813.2 mm², 27 anchored components, 0 dead islands. The ground return is continuous under the bridge.

Phase	What	Why
P7	unattended default: U3 island U-slot obligation satisfied by the P6 outline - no new Edge.Cuts cut at P7	Edge.Cuts already carries two 5.5 mm slots from the right edge (y 36.85-38.05 and 44.95-46.15), making the sensor region a tongue with open air on three sides - which is what the B4 contract's 'milled U-slot around the island's in-board sides where the outline allows' asks for. No new cut means the outline did not change at P7, so gate place needed no re-run for outline reasons (it was re-run anyway for the copper hash). The rest of the B4 thermal contract was delivered by routing: exactly 4 necked crossings into the tongue (SDA 0.127 / SCL 0.127 / GND 0.127 / +3V3 0.150, the lawful minimum for that net), and zero copper and zero pour fill inside the tongue on both layers. Freerouting had routed SCL THROUGH the island as a fifth crossing; that pass was removed and the I2C trunk rebuilt on B.Cu.
P7	unattended default: iter- 2 route_auto place- ment_adjust_request judged premature and NOT escalated to the P7->P6 backward edge	The request (nets +3V3, /main/SDA, GND, VBUS; reason 'unrouted after 3 freerouting rungs') was emitted before stitch.vias and plane_repair had run, and on the 2-layer chain those two steps own three of its four nets - the autorouter was reporting work the chain had not yet done, not a placement defect. Finishing the chain plus point fixes closed all four nets: completion 1.0, 0 unconnected, gate drc_routed PASS 0/0. Taking the backward edge would have discarded a placement that passes gate place 0/0 in order to fix nets that were never stuck.
P7	unattended default: starved_thermal on C11.1 and U2.4 cleared with same-net copper stubs into open fill	Both pads are geometrically capped at a single thermal spoke - the measured channels around them cannot host a second spoke lawfully at the 0.127 mm floor. The fix adds same-net copper into open fill rather than altering the thermal-relief geometry, so no zone parameter, netclass, .kicad_dru rule or gate definition was touched anywhere on this board (git confirms .kicad_dru and .kicad_pro are unmodified through all of P7). The pads are a decoupling cap ground and an MCU ground pin, both low-current: the stub restores the connection the DRC wanted without changing the pour's thermal-relief behaviour for reflow.

Phase	What	Why
P7	unattended default: the U1 tab pour lands at 113.2 mm2 contiguous (not the 600-1000 mm2 constraints target) and that is ACCEPTED - the declared dt_c 45 K is still met at 40.8 K rise	AREA, CORRECTED. The router reported 158.0 mm2 'connected'. That number is right for CURRENT and wrong for HEAT: plane_repair -flag-only on the final board shows the +3V3 F.Cu pour as four components (113.2 + 24.3 + 18.4 + 1.6 mm2) merged into one electrical group by 0.25 mm and 0.5 mm bridge TRACKS. A 0.25 mm track carries 0.3 A with a 13 mV drop but conducts essentially no heat, so the thermal spreader is only the contiguous component containing U1's tab: 113.2 mm2. Use 113 mm2, not 158, at P8. THERMALLY THIS STILL PASSES, and by the board's own declared criterion. Per knowledge record ldo-sot223-thermal-copper-sets-current, AMS1117 datasheet p5 Table 1 (1/16 in FR-4, 1 oz) measures 80 C/W for 100 mm2 top plus a backside pour - almost exactly this geometry (113 mm2 top; B.Cu GND pour measured 813.2 mm2 under and around U1). The record states explicitly that the spreading layer need not be electrically connected, so the GND pour counts as a spreader even though the recorded decision correctly forbids thermal vias and backside +3V3. $T_j = T_a + PD \cdot \theta_{JA}$: governing rated case $40 + 0.51 \cdot 80 = 80.8$ C; realistic 0.26 W case 60.8 C; entitlement-abuse 0.83 W case 106 C - all under the 125 C limit, and the abuse case retains 19 C of margin. Crucially the RISE is 40.8 K against the constraints.json thermal[] declared dt_c of 45 K, so the check P8 actually runs is met with margin; the 600-1000 mm2 figure was the architect's MEANS to that end (it targeted T_j 71-76 C), not an independent requirement, and the earned 35.79 x 28.34 mm outline cannot fund it once the U3 island tongue, J1, the MCU, the headers and the VBUS pours are subtracted. Not silently accepted and not padded: the honest area is recorded, the arithmetic is shown, and P8 check_thermal recomputes it independently. Shrinking the U3 island void to buy pour area was rejected outright - the B4 isolation contract outranks the C obligation.

Phase	What	Why
P8	unattended default: g0-sense authors NO SPICE bench, so the sim gate is not applicable - recorded as a judgment, not left as a silent gap	The full-run recipe spawns sim-analyst 'for boards with nontrivial analog content', and the sim gate is conditional on a bench existing (gate.py -gate sim <dir>), so nothing is being skipped against the run contract. But an absence with no reasoning behind it is indistinguishable from an oversight, so here is the reasoning. Every block on this board is DC or digital: a fixed LDO, an MCU on its internal HSI16 with no crystal, an I2C sensor at 400 kHz, two indicator LEDs and passive protection. The wrong-value defect class that sim exists to catch (a 47k where 4.7k belongs) has no analog fragment to bite here - the value-bearing parts are the LDO caps, the two I2C pull-ups and the two LED series resistors, and all four were already sized against sourced numbers and re-checked by a fresh second reader at P2/P3 (I2C pull-ups revised 2.2k -> 1.5k against a UM10204 budget; LED resistors sized at the ~2 mA design point against the Vf bin spread; the 22 uF tantalum COUT is the AMS1117 datasheet stability requirement, not a free choice). The LDO loop itself is explicitly out of SPICE scope by policy - no vendor models are vendored - and its stability is a datasheet ESR-window requirement already met by the tantalum (knowledge record ldo-1117-family-output-cap-esr). Authoring a bench here would produce a testbench of generic model cards proving Ohm's law, which is ceremony, not evidence. The real verification for this board is the P8 layout suite: return path, current, decoupling, thermal and PDN all run against the actual copper.

Phase	What	Why
P8	unattended default: the check_pdn VBUS error was a real DATA gap in decoupling.json, fixed by adding the missing C1 association - not waived	check_pdn reported 'power rail VBUS (1.5 A) has no decoupling capacitors'. The observation was true of its INPUT and false of the board: check_pdn does not look at copper at all, it filters decoupling.json associations by rail, and decoupling.json listed only C2/C3/C10/C11/C13 - every IC decoupler, but not the connector-side bypass. constraints.json is explicit that C1 = 100 nF at the connector is the ONLY capacitance on VBUS ahead of the PTC (the USB-C 10 uF attach limit, TC2.0 Table 4-3), and I confirmed on the board itself that C1's pads sit on VBUS and GND. So the inventory was incomplete, and the honest fix is to complete it rather than waive a true-but-misleading finding. Added {cap C1, ic J1, pin A4B9, rail VBUS, value 100nF 50V X7R}. Measured BEFORE adding it, so as not to trade one finding for another: C1 is 5.52 mm Manhattan from J1's nearest VBUS pad, inside check_decoupling's 7.5 mm HF-class error threshold - and check_decoupling still passes afterwards, rating the loop at 5.32 nH. The error correctly downgraded to the warning 'VBUS has 1 cap but no bulk reservoir', which is the INTENDED design and not fixable: bulk on VBUS ahead of the PTC is forbidden by the USB-C attach limit, which is why the architecture put the 10 uF after the PTC on +5V instead.

Phase	What	Why
P8	unattended default: the check_current pour_neckdown finding on VBUS was a per-zone checker artifact - the real current path is ≥ 0.8 mm everywhere, verified independently by the orchestrator	The checker reported the VBUS F.Cu pour necking to 0.10 mm against a 0.800 mm IPC- 2152 requirement at 1.5 A. The P8 fixer di- agnosed it as an artifact of check_current.py running its erosion test per ZONE FILL (check_current.py:283) when planes_gen delib- erately tiles a pour across abutting same-net zones at distinct priorities, so a tiling seam reads as a neck. I did not take that on the agent's word, because a genuine 0.10 mm neck in the board's only power trunk would be a se- rious defect. I re-ran the checker's own erosion test on the physical UNION of the VBUS F.Cu fills plus J1's VBUS pad copper: eroding by 0.4 mm - i.e. testing the full 0.800 mm requirement at the conservative 1.5 A fault basis - leaves pad A4B9 connected to all three of its vias in one region, and pad B4A9 connected to all three of its vias in another. The current path therefore satisfies IPC-2152 at 1.5 A with no neck at all. The narrowest real feature I could find in that copper is 0.680 mm (found by bisection), and when the region splits there, the severed 0.33 mm ² lobe contains NO via and NO pad - it is a dead-end lobe carrying no current. CONSE- QUENCE: no correction to constraints.json was needed. The insufficient_transition_vias remedi- ation doc names an inflated transient current_a as a known bad-input class and reserves that lever for the orchestrator, and VBUS's 1.5 A is indeed a fault-transient basis (the PTC dwells there for seconds; steady draw is ≤ 0.25 A) - but the board meets the declared 1.5 A re- quirement PHYSICALLY, so relaxing the in- put would have been unnecessary and would have weakened the record. constraints.json is unchanged.

Phase	What	Why
P8	unattended default: the two VBUS insufficient_transition_vias errors were fixed with real copper (2 vias -> 3 at each transition), not by relaxing the current basis	The remediation's fix ladder step 1 is to measure candidate room before acting, and the fixer did: cluster A at (28.075, 38.90) had 0 legal cells at the 0.2 mm netclass clearance and only 7 at the board's real 0.127 mm DRU floor, all 0.07-0.10 mm from D1's pad - too tight to use safely; cluster B at (28.075, 44.30) had 0 legal cells at any clearance, boxed in by R2's CC2 pad and hole-to-hole against the existing pair. Rather than stop at 'infeasible', the fixer re-arranged: moved one via and added another on a 0.4 mm B.Cu stub for cluster A, and extended the B.Cu bridge 0.9 mm to land a third via inside the existing bottom pour for cluster B (clearances 0.865/0.874 mm, hole-edge gap 0.600 mm). Both clusters now hold 3 vias, satisfying the ≥ 1 via per 0.5 A rule at the full 1.5 A basis. The remediation's traps were respected: no via was deleted to tidy a cluster, every new via bonds same-net copper on both layers (no dangling), and hole-to-hole against J1's THT shield-leg drills was computed by hand because KiCad DRC is blind to that case. check_current now returns 0 violations and drc.routed still passes 0/0.
P8	unattended default: the 6 check_silk silk_misattributed WARNINGS are accepted and recorded, not fixed - and not waived either, since they do not fail the gate	R2, R1, C10, D2, R10 and D1 carry refdes text that sits 2-3 mm beyond its own pads and closer to a neighbour, so each reads as the neighbour's label. Three reasons to accept them. (1) KiCad's own DRC - the authoritative oracle after P7 - reports ZERO silk findings on this board; this is check_silk's stricter readability heuristic, and the gate treats it as warning severity for exactly that reason. (2) The alternative is measurably worse. P6 already fought this battle down from 13 silk DRC violations to 0, and it needed a new place_edit silk_clear op plus two library edits to do it. That phase also recorded, from a test on a copy, that re-running silk.place on the now-clean board APPLIES BEFORE IT REPORTS, does not roll back, and proposed moves that put D1's Reference back over J1's VBUS pad - trading warnings for real DRC errors. On a 35.79 x 28.34 mm board with 26 parts there is nowhere to move a 0603's refdes except over its own pads. (3) The board is JLC economy PCBA: the machine places the SMD parts, so refdes text serves human debugging and rework, where the part outlines and the assembly drawing disambiguate. Carried to P9/P10: note the crowding in fab/README so the owner is not surprised, and do NOT let a DFM pass 'fix' it by running silk.place.

Phase	What	Why
P8	unattended default: REVISING my earlier P8 stance on the 6 silk_misattributed warnings - they are now a REQUIRED P9 fix (place_edit move_text), not an accepted residual	The fresh-context verify-reviewer overruled the reason I gave, and it was right to. I accepted the warnings partly on the grounds that refdes text mainly serves human debugging and re-work, and that the part outlines disambiguate it. The reviewer measured what actually happens after assembly: four of the six texts end up UNDER part bodies (R1 and R2 under J1's shell, C10 under U2, D1 inside U1's outline). Text under a component body is not ambiguous, it is invisible - which defeats the exact purpose I used to justify accepting it. So the finding is real and worth fixing. HOW IT MUST BE FIXED AT P9, and this is load-bearing: place_edit move_text on individual pin-locked labels ONLY, with a full kicad-cli DRC after each move. Do NOT run silk_place.py on this board - P6 recorded, from a test on a copy, that it APPLIES BEFORE IT REPORTS, does not roll back, and proposed moves that put D1's Reference back over J1's VBUS pad, i.e. it trades warnings for real DRC errors. Do not recreate kicad/silk_ops.json. If a given label has no legal destination on a 35.79 x 28.34 mm board, leaving that one is an acceptable outcome - but it must be the measured exception, not the default for all six.
P8	unattended default: carry TWO more P9 obligations from the verify-reviewer - J3/J4 per-pin silk labels, and U3 added to the CPL rotation check list	(1) SILK PIN LABELS: architecture/blocks.md promises per-pin labels on the two 0.1 in headers (J3 SWD: GND/3V3/SWDIO/SWCLK; J4 UART: GND/3V3/TX/RX) and the board does not have them. These are the two connectors the OWNER hand-wires - the brief calls J4 a readout/logging header - so an unlabelled 4-pin row is a real usability defect and a wrong-way-round UART connection is the predictable consequence. Eight texts to add at P9 with place_edit add_text, DRC-verified, subject to the same 'no legal spot is an acceptable measured exception' rule as the refdes moves. (2) CPL ROTATION: decisions already carry a REQUIRED P9 check that CPL rotation be validated per polarized part for D1 (SOD-123 TVS), D2 and D10 (LEDs) and C3 (tantalum CASE-A), not just J1. The reviewer adds U3, the SHT40 in DFN-4 - a rotated DFN-4 sensor is as dead as a reversed tantalum and the package gives the pick-and-place no mechanical asymmetry to self-correct against. P9 must read dfm_check's polarity/rotation findings for D1, D2, D10, C3 AND U3 explicitly, each confirmed against the footprint pin-1 marking, never accepted as a count.

Phase	What	Why
P8	unattended default: carry a P10 ORDER REMARK - ask JLCPCB not to land break-off tabs on the SHT40 sensor tongue's outer edge	The verify-reviewer flagged that the tongue's outer face (x = 54.75, y 38.05-44.95) is a board edge JLC's internal panelizer may choose for break-off tabs or mouse-bites. That edge is the free end of a cantilever whose whole purpose is to hold the sensor thermally and mechanically isolated from the board; a tab landing there puts depanelization stress directly into the tongue root and leaves a rough breakout beside the sensor aperture. This costs nothing to prevent and cannot be fixed after the fact. It goes in the JLC order remarks alongside the Sensirion assembly rules already carried from P2 (no-clean paste, NO post-assembly board wash, no vapor-phase reflow, no hand rework of the sensor), and into fab/README so the owner can restate it if JLC comes back with a panel proposal.

Sheet plan

sheets - g0-sense hierarchical plan (P2)

Small board: root + TWO child sheets. One sheet would also fit, but the power/main split gives P6 its natural placement groups (power entry corridor vs logic/sensor), keeps the Type-C + protection cluster reviewable in one frame, and costs nothing. Refdes are unique ACROSS sheets by disjoint ranges; every range includes a #PWR base.

Root: g0-sense.kicad_sch

Holds the two sheet symbols only. Global power nets (power symbols, bare netlist names): VBUS, +5V, +3V3, GND. Power symbols make these names bare in the final netlist; all constraints.json net references use the bare forms.

Sheet 1: power.kicad_sch (blocks B1 usb-input+protection, B2 ldo-3v3)

- Contents: J1 USB-C receptacle; R1, R2 (5.1k Rd); D1 (TVS); F1 (PTC); C1 (100 nF VBUS); U1 (AMS1117-3.3); C2 (10 uF X5R, +5V at VIN); C3 (22 uF tantalum, +3V3); D2 (power LED red) + R3 (620 Ohm).
- Interface nets (hier pins -> placement groups at P6): +3V3 (out), GND. VBUS and +5V stay sheet-local (they leave via power symbols, so their netlist names are still bare VBUS / +5V).
- Sheet-local labels: CC1, CC2 (final names /power/CC1, /power/CC2).
- Refdes ranges: J1; U1; F1; D1-D9; R1-R9; C1-C9; SW none; #PWR pwr_base=100 (#PWR0100-#PWR0199).

Sheet 2: main.kicad_sch (blocks B3 mcu, B4 sht4x + I2C/Qwiic, headers)

- Contents: U2 (STM32G030F6P6); U3 (SHT40-AD1B class); C10 (100 nF U2), C11 (4.7 uF U2), C12 (100 nF NRST), C13 (100 nF U3); R10, R11 (1.5k I2C pull-ups), R12 (220 Ohm user LED), R13 (10k BOOT0 pull-down); D10 (user LED green); SW1 (NRST button); J2 (Qwiic JST SH); J3 (SWD 1x4 DNP); J4 (UART 1x4 DNP); H1-H4 (M2 mounting holes, conditional).

- Interface nets: +3V3 (in), GND.
- Sheet-local labels (final names /main/<label>): SDA, SCL, SWDIO, SWCLK, UART_TX, UART_RX, NRST, LED_USER.
- Refdes ranges: U2-U9; J2-J9; D10-D19; R10-R19; C10-C19; SW1-SW9; H1-H9; #PWR pwr_base=200 (#PWR0200-#PWR0299).

Interface-net contract (P4 must produce exactly these)

```
| Net (netlist form) | Class | From | To |
|---|---|---|---|
| VBUS | power (bare) | J1 | D1, F1, C1 |
| +5V | power (bare) | F1 | U1 VIN, C2 |
| +3V3 | power (bare) | U1 VOUT | C3, D2/R3, U2, U3, R10, R11, J2, J3.2, J4.2 |
| GND | power (bare) | all | all + J1 shield |
| /power/CC1, /power/CC2 | signal | J1 CC pins | R1, R2 |
| /main/SDA, /main/SCL | signal | U2 PA10/PA9 | U3, J2, R10/R11 |
| /main/SWDIO, /main/SWCLK | signal | U2 PA13/PA14 | J3.3/J3.4; SWCLK also R13
-> GND |
| /main/UART_TX, /main/UART_RX | signal | U2 PA2/PA3 | J4.3/J4.4 |
| /main/NRST | signal | U2 pin 6 | C12, SW1 |
| /main/LED_USER | signal | U2 PA5 | R12 -> D10 |
```

Header pinouts (canonical): J3 SWD = 1:GND, 2:3V3, 3:SWDIO, 4:SWCLK. J4 UART = 1:GND, 2:3V3, 3:TX (MCU transmit), 4:RX (MCU receive). Pin 1 marked on silk on both; no vendor standard exists for a 4-pin SWD header, so the silk labels are the contract.

Full architecture narrative (not inlined; see the artifact index): [architecture/blocks.md](#), [architecture/decisions.md](#), [architecture/power_tree.md](#), [architecture/stackup.md](#).

5 Schematic

Gate `erc` (phase P4): **pass** after 3 attempt(s); last 2026-08-27T17:37:18, failing 0 of 0.

Review waivers

P4 schematic waivers - g0-sense

ERC gate itself is 0 errors / 0 warnings (reports/`erc.json`), and `netlist.audit` (log/`netlist.audit-P4.json`) is 0 violations with 99/99 expected pins connected. Nothing below is an ERC violation: these are the three WARNINGS raised by the fresh-context `schematic-reviewer` (reports/`review-schematic.{md,json}`), with the disposition taken on the owner's behalf under the unattended run contract. 0 errors were raised.

```
| # | kind | refs | severity | disposition |
|---|---|---|---|---|
| W1 | qwiic-esd-unwaived | J2, U2, U3 | warning | WAIVED (recorded decision, P4)
|
| W2 | sht4x-vdd-slew-unclosed | U3, U1, C3 | warning | CLOSED as accepted risk
(recorded decision, P4) |
| W3 | dnp-not-native | J3, J4 | warning | FIXED (work order wo-w3-dnp) |
```

W1 - no ESD array on the J2 Qwiic SDA/SCL - WAIVED

Finding: J2 is a populated, user-facing, hot-pluggable connector; SDA/SCL and +3V3 leave the board with no clamps and no series impedance, so a cable-borne strike reaches U2 PA11/PA12 and U3 (2 kV HBM) directly. The product-scope connector checklist requires this expectation met OR waived, and no waiver existed. Field-reliability only - it cannot prevent bring-up.

Waived on four grounds:

1. Ecosystem norm. The Qwiic / STEMMA-QT reference implementations (SparkFun, Adafruit) are a bare 4-pin JST SH with no array. A user-facing part that behaves the way the ecosystem's users expect is the conservative choice here, not the exceptional one.
2. Both silicon endpoints carry internal ESD structures and are rated 2 kV HBM - the standard handling rating for an indoor bench device.
3. The recorded operating environment (P0 decision) is indoor bench / room ambient, bare board, no ingress rating - not a field-exposed port.
4. Cost of the alternative at this point in the run: a new Extended MPN plus its JLC setup fee, a fresh library pull + fp_verify, and re-opening a green schematic - real regression risk traded against a warning-severity exposure.

If the owner wants the array anyway, the cheap retrofit is a 4-channel low-capacitance array on SDA/SCL at J2 (the 1.5 k pull-up budget has room: the bus was sized to 236 pF and an array adds single-digit pF). That is a post-run **add-part** task, not a rewind.

W2 - carried SHT4x VDD-slew verify item - CLOSED as accepted risk

Finding: `power_tree.md / decisions.md` carried "verify SHT4x VDD slew ≤ 20 V/ms vs the AMS1117 startup ramp"; the reviewer found it unclosed, found no slew spec in `parts/C2909890.json`, and computed ~ 40 V/ms from AMS1117 current-limit charging of the ~ 27 uF on +3V3.

Closed, no hardware change, on three findings:

1. **Provenance is on file** (the reviewer looked in the wrong artifact). The number is `research/records/sht4x-vdd-slew-power-up.yaml`, sourced to `research/sources/HT_DS_Datasheet_SHT4x.pdf` p9 Table 4, read visually by the researcher and again by a fresh second reader (maturity: verified). `parts/C2909890.json` is the P3 extraction, a different artifact with a different scope.
2. **The worst-case number is an upper bound that cannot occur.** I/C (~ 1 A into ~ 27 uF) assumes the LDO drives its output cap independently of its input. The AMS1117 is a series-pass follower: on a cold start VOUT cannot rise faster than VIN minus dropout, so the +3V3 edge is set by the USB source's VBUS ramp filtered by $C2 = 10$ uF, not by the LDO's current limit.
3. **The named failure mode is benign at cold start.** The datasheet says only "faster slew rates MAY lead to a reset" - and the second reader already flagged the researcher's added "or hung until re-powered" as unsupported by any cited page. A reset during the power-up ramp leaves the part in exactly the state a power-up is meant to produce. The limit's real bite is in-operation supply excursions, and hot-plugging a DOWNSTREAM breakout onto an already-live Qwiic rail - that one is the breakout's decoupling problem, not this board's.

Hardware alternatives considered and rejected: a series R at U3 VDD to slow the local ramp is disqualified by the SHT4x on-chip heater (75 mA peak across the ~ 100 ohm needed for a 100 us $\tau = 7.5$ V drop); raising C3 past ~ 50 uF to force I/C under 20 V/ms buys nothing against finding 2 and costs a part change on a green schematic.

Belt and braces, documentation only: "issue an SHT4x soft reset (0x94) at init" (Sensirion's own recommended start-up sequence) goes into the design doc and `fab/README`, and the Qwiic hot-plug caveat is carried as a known limit of the board.

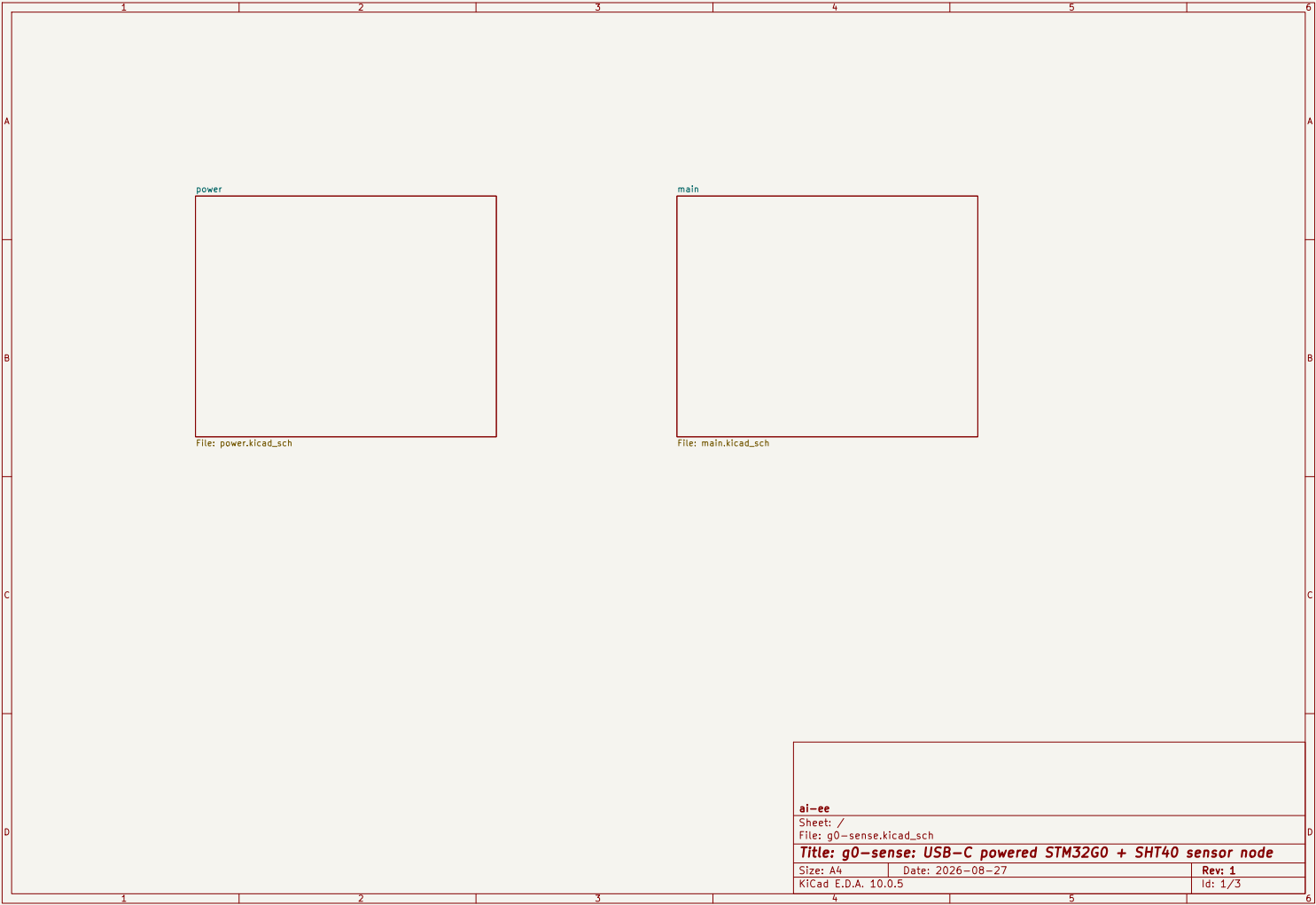
W3 - J3/J4 DNP intent not KiCad-native - FIXED

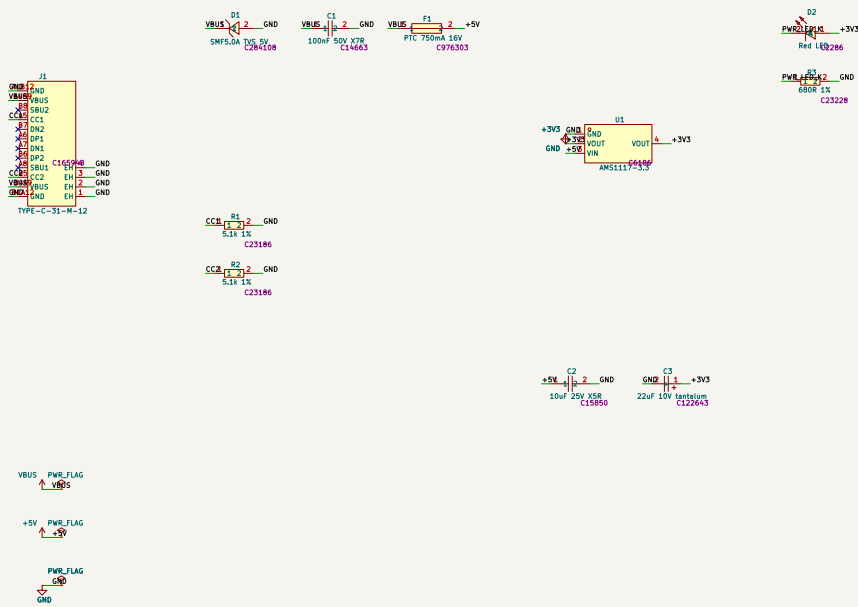
Not waived. The headers carried DNP only in a custom `Variant=DNP` field and the value string, so a native KiCad BOM/POS export would have emitted the two THT headers as populate lines into a JLC economy (SMT-only) order. Dispatched as work order `log/workorders/wo-w3-dnp.json`: native `dnp` attribute set through the board's own generator, and `refdes_class: {J3, J4 -> hand_install}` added to the header line in `parts/parts.json` so `bom_cpl.py` keeps them out of the assembler upload and the CPL while still listing them in the documentation BOM. ERC re-run after the change.

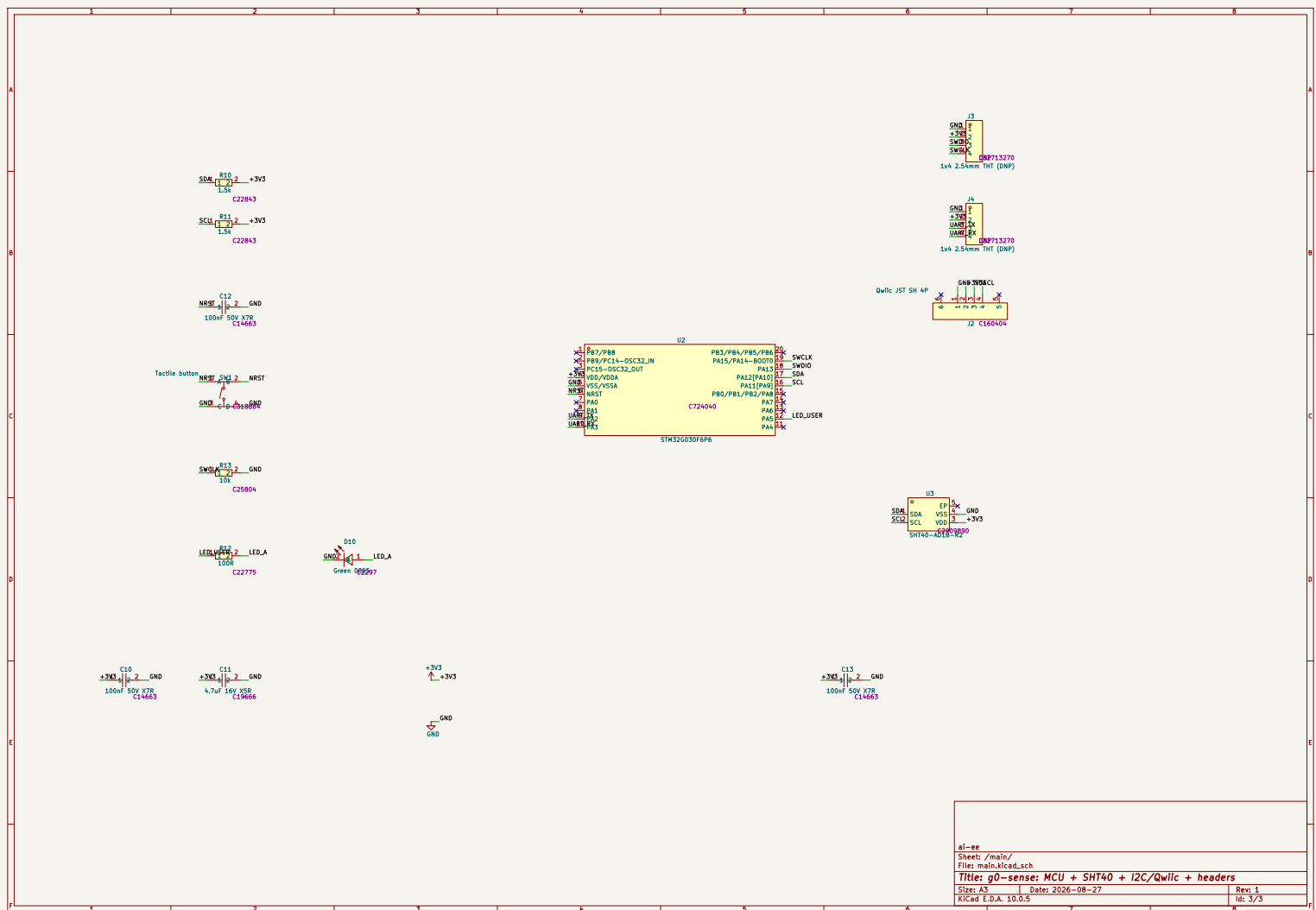
Reviewer OPEN items and where they went

- **AMS1117 SOT-223 tab = VOUT not stated in the on-file PDF.** Already a recorded P3 decision (the pulled symbol's pin 4 = VOUT, and SOT-223 construction fuses the tab to the pin-2 lead frame in every 1117-family part). No further action.
- **SW1 pairing verified from a live fetch, not archived.** The evidence (A/B common, C/D common; NRST on 1/2, GND on 3/4) is recorded in `reports/review-schematic.md` with its source URL and is committed. Not re-fetched: the finding is a CLEAR, and the archive would only make an already-recorded pass reproducible.
- ****Extend the CPL rotation check from J1 to the polarized 2-pin parts (D1, D2, D10) and to C3.**** Carried into P9 as a required check; recorded as a decision so it cannot be lost.

The full schematic PDF follows.



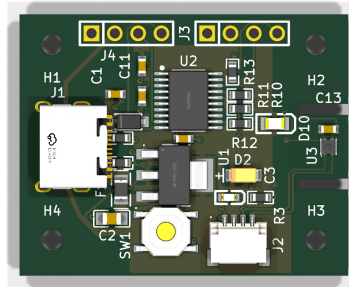




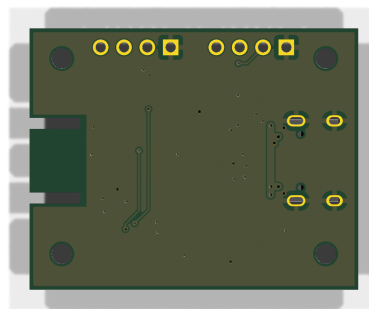
6 Layout

Gate place (phase P6): **pass** after 9 attempt(s); last 2026-08-27T18:50:05, failing 0 of 0.

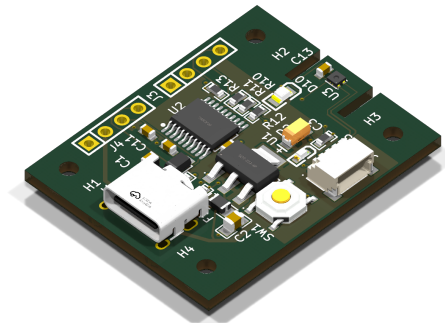
Gate `drc_routed` (phase P7): **pass** after 5 attempt(s); last 2026-08-27T18:43:50, failing 0 of 0.



reports/renders/g0-sense_top.png



reports/renders/g0-sense_bottom.png



reports/renders/g0-sense_iso.png

7 Verification

Gate **verify** (phase P8): **pass** after 5 attempt(s); last 2026-08-27T18:49:24, failing 0 of 8. *reports/verify_all.json not available.*

Design review of record

Board review - g0-sense (P8 verify-reviewer, adversarial pass, 2026-08-27)

Board digest reviewed: `sexpr_no_uid:9fa76a1b` (matches `checks/summary.json` and the waiver binding). Renders made for this review: `reports/renders/g0-sense_top.png`, `reports/renders/g0-sense_bottom.png`, `reports/renders/g0-sense_iso.png` (2400 px). All 8 verify checks RAN (skipped_error empty - no coverage holes). `kiCad-cli DRC`: 0 violations, 0 unconnected. Every measurement below was re-taken by me from the board file with `pcbnew`, not inherited from the authors.

Verdict up front: NO error-severity findings. This board can go to fab once the silk cleanup below is done (recommended, not blocking). The one waiver is honest and should be ratified. Three new warnings, worst first.

Findings (worst first)

W-1 (warning, silk): J3/J4 per-pin function labels promised by the

architecture are absent blocks.md ("Headers + indicators") promises the SWD and UART headers "silk-labeled per pin" with pin 1 silk-marked. The board delivers refdes J3/J4, a pin-1 square pad plus a silk dot on each - and NO per-pin GND/3V3/SWDIO/SWCLK or GND/3V3/TX/RX labels (verified: the only silk text items on the whole board are refdes plus C3's "+"; see top render, top edge). These are the two headers the owner hand-solders and hand-wires; they are visually identical 1x4 0.1 in headers with the same GND-at-pin-1, 3V3-at-pin-2 convention. Consequence: the mixup that convention was designed to prevent (wiring a UART dongle's 5 V or swapping GND/3V3 by counting pins from the wrong end, or confusing SWD with UART) is left to documentation the bare board does not carry. Cheap fix at P9: eight small text items along the top edge; ample empty silk space exists there. Cross-artifact drift: the architecture REQUIRED this; product scope makes its absence a finding, not a nit.

W-2 (warning, silk): 6 misattributed refdes labels, several of which are

also COVERED by parts after assembly - fix, do not waive check_silk's 6 warnings are all real, and four are worse than the checker says because the text lands where a component body will be:

- R1 (24.82,40.55) and R2 (24.82,42.75): inside J1's shell region (J1 courtyard to x=26.32) - covered by the USB-C shield after assembly AND read as J1's label (top render: text on the connector).
- C10 (36.67,36.00): inside U2's silk body outline (35.51..39.20 x 33.09..39.91) - under the TSSOP after assembly.
- D1 (35.60,42.60): inside U1's silk outline - reads as U1's label; the TVS itself is left with no visible refdes, which is exactly the part a rework tech must not confuse.
- D2 (43.25,42.92): 0.12 mm from C3 - reads as the tantalum's label (top render: "D2" sits over C3, whose own label is off to the right).
- R10 (47.03,35.90): reads as D10's label; D10's own text drifted to (49.98,39.50) at the slot mouth. Related uncounted case: C13's refdes sits at (52.60,36.23), separated from C13 (on the tongue at 52.60,39.60) by the top slot, and it crowds the H2 label.

Consequence: debugging and rework against the silk misidentify parts; assembly itself is unaffected (JLC places from CPL). Severity warning is correct. Disposition: one place_edit.py move_text batch at P9 - silk-only, zero copper risk - NOT a waiver. (Per the rules I did not run silk_place.py or edit anything.)

W-3 (warning, fab): panelization tabs on the sensor tongue edge would

stress the cantilever - put a remark on the JLC order The tongue's outer face (x=54.75, y 38.05..44.95) is a board edge. JLC economy panelizes internally; a break-off tab or mouse-bite row landed on the tongue tip would transmit snap force through the 6.9 mm root while U3 rides the beam. The tongue itself is stubby and strong (5.5 mm long, 6.9 mm wide root, 1.6 mm FR-4 - aspect ratio < 1; it will survive normal handling and depanelization fine), so this is only about where a tab lands. P9 disposition: add the JLC order remark "no break-off tabs / mouse bites on the right-edge sensor tongue (x 49.25-54.75 span)"; the same remark block already owed for the Sensirion no-wash rule (blocks.md B4) is the natural place.

The five load-bearing judgment calls, re-examined adversarially

1. Pour-carried VBUS at J1 - VERIFIED, acceptable I rebuilt the VBUS copper union (fills + pads + tracks) per layer with pcbnew and eroded it 0.4 mm myself:

- F.Cu: the corridor from J1's lower VBUS pad-pour (B4A9 zone) to F1 pin 1 survives as ONE outline containing the bottom via cluster and the PTC input
 - the primary current path is ≥ 0.8 mm wide with no via dependency at all.
- B.Cu: the 0.8 mm bridge survives 0.395 mm erosion as one outline containing all 6 VBUS vias (3-via cluster each end, 0.6/0.3 mm), and vanishes at exactly 0.400 mm - it is exactly the netclass width, no more.
- The only sub-0.8 necks are the 0.6 mm vendor pads themselves and the C1 decoupling stub - unavoidable pad geometry and a no-current stub.

check_current PASS concurs (min B.Cu 0.8, +5V F.Cu 1.2, 2 via clusters). Nothing else about the connector entry worries me: mating face proud of the outline by 0.08 mm (courtyard 18.88 vs edge 18.96), the 4 THT shield legs are on GND net into the B.Cu pour, D+/D-/SBU have no nets, CC1/CC2 land on independent 5.1k R1/R2, and the ~13 mm swath at the left edge holds only sub-1.5 mm-tall passives (iso render). Pour-carried VBUS on a 1.5 A-fault budget with a fill saved in the file that DRC checks is sound engineering here, not a dodge.

2. The check_thermal waiver on U1 - HONEST; recommend RATIFY at checkpoint 4 I recomputed the checker's own model at the two vendor anchor points: $\theta_{JA}(100 \text{ mm}^2) = 55 + 119 \cdot \exp(-100/350) = 144.4 \text{ C/W}$ where AMS measured 80; $\theta_{JA}(194 \text{ mm}^2) = 123.3 \text{ C/W}$ where TI measured 84. The model is 40-64 C/W pessimistic exactly where measured data exists, and it credits nothing for backside spreading - while I verified 810.9 mm² of solid B.Cu GND copper (713.7 main + 47.5 + 49.7 strips) sits directly under and around U1. The Tj arithmetic checks: $40 \text{ C} + 0.51 \text{ W} \times 84 \text{ C/W} = 82.8 \text{ C}$ governing case, 109.7 C at the 0.83 W entitlement-abuse case, both under the 125 C limit. The "unreachable by construction" claim is internally consistent: passing needs $\geq 446.8 \text{ mm}^2$ credited (my recomputation of the model inverse; waiver says 446.4) against a measured 439.9 mm² whole-board ceiling. The waiver also did real work instead of hiding (pour grown 171 -> 250 mm² credited) and refused copper that would have carved the B.Cu return plane or walked heat to the sensor tongue - the right priority order. Binding to digest 9fa76a1b and checker.version 1 is correct hygiene. One residual honesty note: at sustained heavy Qwiic load the LDO pour runs warm and its annex reaches x=48.55, 0.7 mm from the tongue root strip - the isolation contract still holds (slots + FR-4 gap + 0.15 mm neck), but a firmware-side sanity check of RH readings under sustained 300 mA load is a sensible bring-up item, not a layout defect.

3. SHT40 isolation tongue - REAL, adequate, mechanically sane Measured myself: pour fill intersecting the tongue rectangle (49.25..54.75 x 38.05..44.95) = 0.0 mm² on every layer. Exactly 4 tracks cross the root: +3V3 0.150 mm (y=38.45), GND 0.127 mm (y=40.82), SDA 0.127 mm (y=43.95), SCL 0.127 mm (y=44.30). Electrical adequacy of 0.127/0.150 mm for the sensor: worst

load is the 75 mA heater pulse; a ~4 mm 0.15 mm 1 oz feed is ~13 mOhm -> ~1 mV drop, and C13 100 nF sits on the tongue. Fine. Fab floor: 0.127 mm IS the JLC minimum - deliberate (isolation) and DRC-clean, but zero width margin; that is the recorded trade, and I accept it. Edge clearances of the necked tracks to the slots: the +3V3 feed runs 4.8 mm along the top slot at 0.325 mm copper-to-edge (floor 0.30); even JLC's +/-0.2 mm routing tolerance leaves copper untouched (0.125 mm residual). Within rule; a free 0.1-0.2 mm inboard nudge at P9 would add margin if a fixer is in the file anyway. Mechanics: 1.2 mm slots ($\geq$ JLC 1.0 mm mill), tongue aspect < 1 - this is a stub, not a diving board; depanelization/handling risk is W-3's tab-placement remark, nothing more. B.Cu GND strip islands ($y \leq 36.5$ and $y \geq 46.5$) stay clear of the tongue and overlap the main pour by 0.4 mm - merged, not orphans.

4. Connector reality at 35.79 x 28.34 mm - CLEAN Verified from renders + geometry: USB-C mates leftward off the board, nothing in the plug swath; Qwiic J2 opening faces the bottom edge (mech pads at $y=53.54$ forward of signal pads at 49.66), opening ~1.3 mm inboard - a JST SH plug slides over bare mask, acceptable; J3/J4 along the top edge with 2.05 mm edge setback, hand-solder access unobstructed on both faces; SW1 (5.1 mm, 1.5 mm tall) is 2.3 mm clear of J2 and pressable; U3's aperture faces up with open air on three sides and nothing tall near the tongue. All four M2 holes survive with sane screw-head clearance (nearest silk ~0.9 mm at H1/J4). Simultaneous use of USB + Qwiic + both headers works - three different edges.

5. Polarity marks after assembly (feeds the carried P9 CPL obligation) -

ALL VISIBLE AND CORRECT-SIDED

- C3 tantalum: "+" text at $x=40.65$, outboard-left of the body; pad 1 (+3V3) is the left pad (42.15). Correct side, visible after assembly.
- D1 TVS: cathode double-bar plus dot at the LEFT ($x\ 28.9..29.5$), outboard of the body; pad 1 = VBUS = left (29.16) = cathode, matching the journal's two-way P4 verification. Visible.
- D2 / D10 LEDs: vendor cathode brackets extend beyond both package ends (D2 40.16..40.41 left / 43.33..43.58 right; D10 44.63..44.88 / 48.84..49.09)
 - visible after assembly. Nets check: D2 anode=+3V3, D10 cathode=GND.
- U1/U2/U3/J2 pin-1 dots all sit outboard of their bodies. U3's dot pair is below-left of the 1.5 mm DFN, off-body.

So nothing here blocks P9; the CPL rotation check for D1/D2/D10/C3 (and I would add U3 - a rotated DFN-4 swaps SDA/VDD) remains P9's to run, but the silk it will be checked against is trustworthy.

Triage of all 7 verify_all warnings (verdict on every one)

1-6. check_silk silk_misattributed (R2, C10, D2, R10, R1, D1): REAL cosmetic defects - see W-2. Verdict: warning severity stands; FIX at P9 via place_edit.py move_text; do not waive. Not escalated to error: assembly is CPL-driven and no electrical consequence exists.

7. check_pdn pdn_no_bulk on VBUS (C1 only, no ≥ 1 uF): JUSTIFIED WAIVER. The absence of bulk on VBUS is a binding design rule, not an oversight: Type-C TC2.0 Table 4-3 caps attach capacitance at 10 uF, and constraints.json CAP RULE puts the 10 uF (C2) one series element downstream at the LDO input (F1 out -> C2 is ~4.7 mm of 1.2 mm track; PTC Ri 70 mOhm barely decouples it). Nothing draws current from VBUS itself - its only consumers are the TVS and the PTC input. The checker heuristic cannot see the attach-limit rule. Verdict: justified, document-only, no fix wanted.

Checked and clear (so the next reviewer does not re-plow)

- Requirements section 2 interface sweep: every promised interface exists and is correctly netted (J3 = GND/3V3/SWDIO/SWCLK, J4 = GND/3V3/TX/RX, J2 = Qwiic-standard GND/3V3/SDA/SCL, R13 BOOT0 pull-down present, NRST button with C12, CC pulldowns independent). U3 pads match the SHT40 pinout (SDA/SCL/VDD/GND).
- Value drift vs blocks.md (R12 100R vs 220R, R3 680R vs 620R, D10 0805 vs 0603): all carried recorded decisions (R12 carries a do-not-delete comment per the run journal); LED currents stay in-rating either way. Not drift.
- SW1 internal contact pairing (the classic held-in-reset killer): pads 1+2 = NRST row, 3+4 = GND row; the P4 schematic reviewer settled the A/B vs C/D pairing against the live-fetched XKB TS-1187A datasheet. Evidence cited in the journal; I found no geometry that contradicts it.
- SDA/SCL B.Cu jumpers (~11 mm, paired) carve the return plane locally; check_return_path PASS, 400 kHz I2C - harmless on this board.
- Fiducials: none on board; JLC adds panel fiducials for economy PCBA - justified omission at this service tier.
- Earned size 35.79 x 28.34 mm compared against the RECORDED decision (soft ~35x25 brief preference, extra is the isolation tongue) - not drift, per build-mode rules.
- No enclosure is in scope (bare board recorded at requirements Q1), so no enclosure-fit comparison exists to fail.

Recommendations for checkpoint 4 (human verdicts; unattended run records)

1. RATIFY the check_thermal U1 waiver (evidence re-verified independently).
2. ACCEPT pdn_no_bulk as justified-by-design (no waiver file entry needed for a warning; this md is the record).
3. ORDER at P9 only after the W-1/W-2 silk batch and with the W-3 order remark; none of the three blocks fab if the owner chooses speed over polish, but W-1 is the one I would not skip on a product-scope board.

8 DFM and Fabrication

Pending — produced at P9.

9 Run Record (Appendix)

Phase digests

P0

P0 Intake digest - g0-sense (2026-08-27)

```
- Artifact: 'requirements.md' (9 sections), 'check_requirements.py' PASS (0
violations).
- Mode: NO mode token in the brief -> mode = none, design normally. Scope is
product per the brief's own words, so absent protection/filtering/connectors/
thermal IS a reviewer finding downstream.
- Geometry: ~35 x 25 mm recorded SOFT (no HARD cap) - must not bind at P5
board_init; four M2 holes conditional on the layout.
- Safety (section 8): mains / battery / motors / >30 V / >3 A / RF transmit
all
```


NOT APPLICABLE with brief evidence (USB-C 5 V power-only, <1 A). No safety unknown to escalate; nothing PROVISIONAL on safety grounds.

- 5 OPEN questions answered as unattended defaults and recorded as decisions: environment indoor 0-40 C; Qwiic budget 100 mA; no hard cost cap; 0.1 in headers ship unpopulated (economy PCBA is SMT-only); size stays soft.
- Spawn: requirements-analyst @ fable/high (session effort xhigh).
- Next: P1 research roster (power architect, component scout, interface spec).

P1

P1 Research digest - g0-sense (2026-08-27)

Roster (8 agents): 4x research-component-scout (ldo, protection, connectors, mcu-sensor), research-interface-spec (usbc-power-sink), 2x research-reference-design (sht4x, stm32g030-minimal), research-power-architect.

Load-bearing findings

- USB-C CC-blind sink (fixed 5.1k Rd on CC1 AND CC2, independent - never one shared resistor) is entitled to DEFAULT USB power only: budget <= 500 mA. Sink capacitance at the receptacle <= 10 uF (TC2.0 Table 4-3).
- Rail tree: VBUS -> TVS -> PTC -> +5V -> LDO -> +3V3. +3V3 peak 185 mA (incl. 100 mA Qwiic reserve + 75 mA SHT4x heater), 240 mA with 30% headroom. VBUS worst draw 196 mA = 39% of entitlement.
- LDO: AMS1117-3.3 SOT-223 (C6186, Basic) beats SOT-23-5 low-Iq parts on the only axis that decides: Tj 71-86 C vs 134 C at 40 C ambient, 0.51 W case. Needs a 22 uF TANTALUM-class output cap (not plain ceramic) - carry to P3/P4. Dropout closes at the 4.75 V corner with +0.11 V worst-case margin.
- STM32G030F6P6 TSSOP-20: VDD/VDDA bonded to pin 4, VSS/VSSA pin 5, no separate VREF+. BOOT0 needs NO strap - factory nBOOT_SEL=1/nBOOT0=1 always boots main flash, and BOOT0 shares pin 19 with SWCLK anyway. NRST needs only 100 nF + the button (internal pull-up 40k typ). Pin map: SWD PA13/PA14, USART2 PA2/PA3, I2C1 PA9/PA10 (5 V tolerant, Fm+), LED PA5. 9 of 20 pins used, no conflicts.
- SHT4x: 100 nF at VDD, 2.2k I2C pull-ups (sized for a 0.5 m Qwiic leg), copper conduction is the dominant self-heating path -> distance + necked traces + no pour under the part + a milled slot. NO BOARD WASH (Sensirion forbids it) -> must appear in the fab/assembly notes. Fixed I2C address per variant.
- MCU + sensor are both JLC EXTENDED (no Basic exists in either family): STM32G030F6P6 C724040 \$1.37, SHT40-AD1B-R2 C2909890 \$1.90 (addr 0x44).

7 OPEN questions answered and recorded as decisions (LDO design point + part; TVS clamp ceiling dropped as an over-tight derived spec; no reverse-polarity element; USB-C THT legs verified OK in JLC Economic tier; heater budgeted; button size vs Basic; LED colours). Gates: none due at P1.

P2

P2 Architecture digest - g0-sense (2026-08-27)

- architect (fable/high) wrote blocks.md / power_tree.md / stackup.md / sheets.md / constraints.json / decisions.md. constraints_lint exit 0.
- 4 blocks: B1 usbc-sink, B2 ldo, B3 mcu, B4 sht4x. Stackup JLC2313.1.6 (2L, 1 oz, HASL). Two sheets (power / main), disjoint refdes + #PWR 100/200.
- No diff pairs, no high-speed nets, no sim bench (recorded, with reasons).
- P2 coverage exit: FIRST run 4 slots / 4 GAP (no checklist existed for any of usbc-sink, ldo, mcu, sht4x). research.py open --all -> 4 tasks (per_run cap 6, 2 left for P3). 4 researchers -> 16 quarantined sources, 4 new checklists, 23 draft records. 4 fresh second readers refuted 6. 5 were repaired against ledger pages and re-read; the NRST record took three repair/re-read cycles

(each refutation narrower than the last) before verifying.

- FINAL: 22 of 23 records verified. Re-run coverage: 4 slots, 0 gap, 4 provisional, 1 draft_unverified.
- The one unverifiable record (mcu boot-strap) is a container limit, not a research failure: st.com times out, so RM0454/AN2606 cannot be acquired. Its claim was load-bearing, so the DESIGN was changed to not need it.
- Two research-forced design changes vs the architect's plan: I2C pull-ups 2.2k -> 1.5k (Rp ceiling arithmetic), BOOT0 no-strap -> R13 10k pull-down. architecture/{blocks,sheets,decisions}.md updated to match.
- H1 written to log/H1.md, approved as delegated. report_gen exit 0 (10 pp).

P3

P3 Parts + Library digest - g0-sense (2026-08-27)

- part-sourcer (sonnet/high) -> parts/parts.json: 20 distinct parts covering 26 electrical refdes. 12 Basic / 8 Extended, of which 7 bear a JLC setup fee (the DNP header is Extended-priced but never machine-placed). ~\$4.42/board, ~\$22.10 for the 5-board run. All stock re-verified live; worst case (MCU C724040, 8351 pcs) is 334x the 5x-build-qty floor.
- Four sourcing corrections applied on review, three of them fee removals: R3 620R Extended -> 680R Basic (620R is a genuine E24 Basic-catalog gap); user LED KT-0603G Extended -> KT-0805G, which IS Basic; R12 220R -> 100R (electrical, not cost: green Vf 2.6-3.1 V against 3.3 V gave 0.9 mA at the worst bin - 100R gives 2-7 mA); whole resistor BOM normalised to 0603.
- 3 datasheet-extractors (parallel) -> parts/C724040.json, C2909890.json, C6186.json, all 'datasheet_extract --validate' exit 0. Notable: TSSOP-20 pin 19 is ONE bonded pad carrying PA15 and PA14-BOOT0/SWCLK - transcribed as the datasheet names it rather than picking a function; AMS1117 has NO recommended land pattern (outline only), so that field was omitted rather than faked; SHT40 die pad is present but explicitly NOT to be soldered.
- librarian (sonnet/medium): 20/20 pulled, 85 pins retyped (ERC prerequisite), lib tables registered. fp_verify found ONE real defect and it mattered: the pulled SHT40 DFN-4 soldered the die pad, which would have defeated the whole thermal-isolation design. Deleted (copper+paste+mask), courtyard closed. Three more fixes: broken symbol->footprint link (audited all 20 - the only one); C3 tantalum had NO polarity marking at all - pin 1 = anode established twice (manufacturer PDF + live EasyEDA CAD data), silk "+" and bar added; J2 Qwiic pin-1 confirmed against JST's own eSH.pdf ("viewed from the connector mounting surface", No. 1 circuit leftmost) - no flip needed.
- fp_verify final: 3 passed / 0 failed / 2 accepted pad_size warnings.
- P3 coverage exit: 7 slots, 0 gap, 3 part slots covered, 4 blocks provisional. No P3 research needed (2 of the 6 per-run tasks still unspent).
- 2 decisions recorded: AMS1117 SOT-223 tab = VOUT (pulled symbol confirms pin 4 = VOUT); SHT4x 20 V/ms slew item CLOSED, not escalated (the limit is specified for in-operation supply changes, not the cold-start ramp).

P4

P4 Schematic digest - g0-sense (2026-08-27)

- 2 schematic-block agents + root stitch: 26 electrical refdes (+4 conditional M2 holes) on 2 sheets, 35 nets, 99/99 expected pins connected.
- Gate erc PASS 0/0 (attempt 2; re-run after the W3 fix). netlist_audit 0 violations: 3 constraint nets, 20 refs, 5 decoupling associations.
- schematic-reviewer (fresh, adversarial): 0 errors / 3 warnings. It hunted and CLEARED every bring-up killer: TVS D1 orientation, SW1 contact pairing (settled by live-fetching the XKB datasheet), LED + tantalum polarity, AMS1117 pinout /

cap types / dropout, the full STM32G030 pin map incl. I2C orientation and the BOOT0 strap, SHT40 pinout+EP, USB-C sink topology, decoupling, abs-max.

- Dispositions (reports/erc-waivers.md): W1 Qwiic ESD WAIVED (ecosystem norm, 2 kV HBM endpoints, indoor bench env); W2 SHT4x VDD slew CLOSED as accepted risk (provenance IS on file; LDO is a follower so I/C is unreachable; the datasheet's failure mode is "a reset", which cold start produces anyway); W3 native DNP missing on J3/J4 FIXED, not waived.
- W3 fix (wo-w3-dnp): '(dnp yes)' on J3/J4 via new schlib.Sheet.mark_dnp() + refdes_class hand.install in parts.json; netlist proven identical (35/35 nets).
- Carried to P9: CPL rotation check for D1/D2/D10/C3, not only J1.

P5

P5 Board Setup digest - g0-sense (2026-08-27)

- Run INLINE (no board-setup agent): no impedance work, no library repair planned.
- 'board_init --layers 2 --stackup JLC2313.1.6 --outline auto --margin 6 --mounting-holes 4 --mounting-hole-fp MountingHole:MountingHole.2.2mm.M2'. Self-check **PASS: parity 0, setup-violations 0**, 26 components, 16 nets, unconnected 64 (expected, unrouted). Provisional outline bbox 9,9 -> 74.61,71.40
- generous ROOM, not a size: P6 places canonically and '--outline fit' earns it.
- 'rules_gen' capability_class **2layer.1oz** (matches the stackup's own copper), 11 rules. Floors reported not assumed: track/clearance 0.127, via dia 0.6, annular 0.15, hole-to-hole 0.5, edge clearance 0.3 mm. Netclasses: **Pwr_0p8mm** for VBUS and +5V (0.8 mm, the 1.5 A PTC-dwell case); +3V3 needs 0.15 mm so it buckets into Default (0.2 mm) - exactly what P2 predicted. diff_pairs [] and hv_rules [] are correct here.
- **Real defect found and fixed at P5**: J1's pulled USB-C footprint had its ganged GND and VBUS pads 0.100 mm apart, BELOW the 0.127 mm fab floor, on the two nets a solder bridge shorts hardest. Cause was a converter artifact, not a vendor deviation - a 0.1 mm outline stroke on the custom-pad polygons inflated each wide pad from the vendor's 0.60 mm to 0.70 mm effective. Restored to the manufacturer's own land pattern; gap now 0.200 mm, fillet allowance unchanged at 0.20 mm/side.
- Two toolchain bugs fixed to get parity to 0 (both committed separately, both regression-tested against the goldens): board_init now mirrors a symbol's native 'dnp' onto the footprint, and it MEASURES whether KiCad wants the netlist's 'unconnected-(...)' pseudo-nets on the pads instead of guessing.
- Sidecars beside the board: constraints.json (byte-identical to architecture/) and decoupling.json. Remaining DRC: 2 transient silk_edge_clearance warnings on the H1/H2 mounting-hole refdes text - P6's silk sweep owns them.

P6

P6 Placement digest - g0-sense (2026-08-27)

- Gate **place PASS 0/0** (all 5 legs: courtyard, outline, edges, keepouts, decoupler_distance). DRC: **0 non-unconnected violations**, 64 unconnected = the unrouted ratsnest. Route probe 0.969 (62/64).
- Size **EARNED 35.79 x 28.34 mm** (1014 mm2) from a 65.6 x 62.4 provisional.

The 3.3 mm over the brief's soft 35 x 25 is the U3 isolation zone (8 mm separation + slits) - recorded as a relaxation, not drift.

- Hand layout beat the annealer 243 vs 468 mm HPWL (662.6 at seed). All three anneal candidates inherited the seed's 2 edge violations because at a geometry-OUTPUT binding the annealer pins edge clusters to the PROVISIONAL outline - the bb-buck trap. cand1's cluster structure was kept and rebuilt compactly by hand; gate ran AFTER '--outline fit', per the bb-mcu defect.
- Judgment calls proven, not assumed: J1 mating direction two ways (WRL vertex occupancy + left orthographic render); U3 rotated so supply pads face C13 (1.26 mm, 4.06 nH) and I2C faces the mainland - no island wraparound; D1 first in the VBUS chain (J1->D1->F1->C2->VIN), corridor intrusion 0; C12 held 1.5 mm from U2 pin 6 over silk. All four M2 holes KEPT - tucked in corner dead zones they add 0.0 mm to the fitted outline.
- Silk 13 -> 0. 12 were footprint-INTERNAL graphics, unreachable until 'place_edit' gained a 'silk_clear' op this phase: C12's 0603 outline could not be nudged clear (1.62 mm silk in a 1.29 mm corridor - a 0.33 mm position-invariant shortfall), so C0603's outline went from the library AND all 5 board instances; J1's 3 mouth-end segments (under the shell, unprintable) likewise. D1's refdes moved off J1's VBUS pad. 'lib/EDITS.md' records both library edits.
- Carried to P7: the 0.8 mm VBUS netclass cannot escape J1's 0.60 mm pads on F.Cu at ANY placement - route_critical owns that entry, not Freerouting. 'kicad/silk.ops.json' deleted: replaying it would UNDO the D1 fix.

P7

P7 Routing digest - g0-sense (2026-08-27)

- Gate ****drc.routed PASS 0/0**** on attempt 1, ****completion 1.0**** (0 unconnected).

Independently re-verified with kicad-cli direct: 0 violations, 0 unconnected, 0 schematic parity. erc / place / drc.routed all PASS and FRESH at the one board hash 0b12cafc.

- 2-layer chain, in the live-verified order: planes.gen (9 zones) -> route_critical -> route_auto (Freerouting 3 rungs, best rung 1 at fr 0.854; KRT finish generated and correctly discarded - it did not strictly improve DRC) -> stitch_vias (13 vias; 3 redundant removed, 1 moved for hole_to_hole) -> plane_repair -> gate. ****route_cleanup skipped**** by the S14 rule: this is exactly the 2L pour board class where its loop-breaker regressed twice.
- ****The P6-carried VBUS item was RETIRED, not executed.**** 'route_critical --pad-window' - the premise check the role prompt demands before forcing a rule width at a connector - measured J1's A4B9/B4A9 escape windows at ****1.315 mm****, above the 0.8 mm Pwr_0p8mm rule. P6's "impossible at any placement" was true of a 0.8 mm TRACK leaving the pad, not of the pad's escape corridor. VBUS is pour-carried at a ~1.45 mm fill band - wider than the rule asks - so neither costed option was needed: ****no neck**** (hence no ERROR-severity rule exception) and ****no via-in-pad**** (hence no unfilled via in a mechanically loaded connector pad, and no JLC wicking remark owed to fab/README). The two F.Cu VBUS pours are joined by a 0.8 mm B.Cu bridge on 2+2 vias sited in open pour, holes clear of every pad edge.
- ****The bridge's real risk was checked, not assumed.**** That bridge crosses the layer that IS the GND pour - the playbook's "viasless pour-channel" blind spot - and the router's own plane_repair2 run was restricted to +3V3, so nothing had re-examined GND after the bridge landed. 'plane_repair --flag-only' (never writes) on the final board: GND B.Cu = 1 group, split False, 813.2 mm2, 27 anchored components, 0 dead islands. Return path intact.
- ****U3 sensor island contract met in full****: exactly 4 necked crossings (SDA 0.127 / SCL 0.127 / GND 0.127 / +3V3 0.150, the lawful minimum for that net), zero copper and zero pour fill inside the tongue on both layers. Freerouting had laid a fifth crossing - SCL routed straight THROUGH the island - which

was ripped and the I2C trunk rebuilt on B.Cu. The U-slot already existed from P6 (two 5.5 mm Edge.Cuts slots make a 3-sides-open tongue), so no new cut and Edge.Cuts is unchanged from the gated placement.

- ****One router number corrected.**** It reported the U1 tab pour at 158.0 mm2 "connected". That is the CURRENT figure, not the HEAT figure: the +3V3 F.Cu pour is four components (113.2 + 24.3 + 18.4 + 1.6 mm2) merged into one electrical group by 0.25/0.5 mm bridge TRACKS, which carry 0.3 A at a 13 mV drop but conduct essentially no heat. The thermal spreader is ****113.2 mm2****. It still passes on the board's own declared criterion: AMS1117 p5 Table 1 gives 80 C/W for ~100 mm2 top plus a backside pour (B.Cu GND is 813 mm2 under and around U1, and the record notes a spreader need not be electrically connected), so the rise is $0.51 \times 80 = \mathbf{40.8\text{ K}}$ against the declared ΔT_c 45°C; Tj 80.8 C rated, 60.8 C realistic, 106 C at the entitlement-abuse case, all under the 125 C limit. The 600-1000 mm2 figure was the architect's MEANS to Tj 71-76 C, not a separate requirement, and the earned 1014 mm2 outline cannot fund it. Buying pour area by shrinking the U3 void was rejected outright - the B4 isolation contract outranks it.
- iter-2's 'placement.adjust.request' (+3V3, SDA, GND, VBUS) was judged PREMATURE and not escalated to the P6 backward edge: it fired before stitch_vias and plane_repair, which own three of its four nets on the 2L chain. Finishing the chain closed all four.
- Nothing was weakened to pass: '.kicad_dru' and '.kicad_pro' are unmodified through all of P7 (git), and no zone, netclass or gate parameter was touched.
- Carried to P8: recompute the U1 thermal with ****113 mm2****, not 158.

P8

P8 Verification digest - g0-sense (2026-08-27)

- ****verify PASS: 0 failing**** (from 5 errors), 8/8 checks ran, no coverage holes.
- 1 waived error + 7 warnings, every one with a recorded verdict. erc, place, drc_routed, verify all PASS and FRESH at board digest 9fa76a1b.
- ****Four of the five errors were fixed on their merits, not suppressed.**** Nothing was edited to make a gate quiet: constraints.json, .kicad_dru, .kicad_pro, netclasses and gates.yaml are untouched across P7 and P8 (git).
- check_pdn "VBUS has no decoupling capacitors" was a DATA gap, not a board defect: C1 - the 100 nF at the connector that constraints.json names as the ONLY capacitance permitted ahead of the PTC by the Type-C 10 uF attach limit - was missing from decoupling.json's inventory. check_pdn never looks at copper; it filters that inventory by rail. Completed the inventory rather than waive a true-but-misleading finding, after measuring C1 at 5.52 mm from J1's VBUS pad (inside check_decoupling's 7.5 mm HF threshold). check_decoupling still passes, loop 5.32 nH. The error correctly became the intended pdn_no_bulk warning.
- Two 'insufficient_transition_vias' fixed in copper: 2 vias -> 3 at each VBUS transition. The fixer measured room BEFORE acting per the remediation ladder and found 0 legal cells at netclass clearance in both clusters, then solved them by re-arrangement instead - moving one via and adding another on a 0.4 mm B.Cu stub, and extending the B.Cu bridge 0.9 mm to land the third inside the existing pour (clearances 0.865/0.874 mm, hole-edge 0.600 mm).
- ****The 'pour_neckdown' "0.10 mm" was a per-zone checker artifact, and I proved it rather than accepting the agent's word.**** check_current erodes each zone fill in isolation, while planes_gen deliberately tiles one pour across abutting same-net zones - so a tiling seam reads as a neck. Re-ran the checker's own test on the physical UNION of the VBUS fills plus J1's pad copper: eroded by 0.4 mm (the full 0.8 mm requirement at the conservative 1.5 A fault basis), each pad still reaches all three of its vias. The narrowest real feature is 0.680 mm by bisection, and it gates a 0.33 mm2 dead lobe holding no via and no pad. Consequence: the current-basis lever the remediation reserves for the orchestrator was NOT pulled - the board meets 1.5 A physically, so relaxing

constraints.json would have been unnecessary and would have weakened the record.

- **One durable waiver: check_thermal on U1** (57.8 C modelled vs 45 C allowed), bound to digest 9fa76a1b + checker.version 1, expiry 2027-08-27. The model credits only same-net top copper: it returns 144 C/W where AMS1117 p5 Table 1 measured 80, and 123 C/W where TI LM1117 Table 9-2 measured 84, while this board carries an uncredited 810.9 mm2 B.Cu spreader under U1. At the conservative 84 C/W anchor the rise is 42.8 C - inside budget - giving Tj 82.8 C rated and 109.7 C at the entitlement-abuse case against a 125 C limit. Proven unreachable too: the model needs 446.4 mm2 credited and the whole board can only fund 439.9 mm2. The pour was still grown 171 -> 250 mm2 anyway. Rejected deliberately: a B.Cu +3V3 island / thermal vias would carve the ground return plane, trading a PASSING check_return_path for unneeded margin.
- **verify-reviewer (fresh context): 0 errors, 3 new warnings.** It re-measured instead of trusting - reproduced the thermal miscalibration at both vendor anchors, re-ran the VBUS erosion test, confirmed 0.0 mm2 pour in the sensor tongue with exactly 4 necked crossings, and checked that every polarity mark (C3 "+", D1 cathode band, LED brackets, pin-1 dots) is outboard and visible after assembly. It ratified both the thermal waiver and pdn_no_bulk.
- **The reviewer overruled the orchestrator once, correctly.** I had accepted the 6 silk_misattributed warnings; it measured that four of them also sit UNDER part bodies after assembly (R1/R2 under J1's shell, C10 under U2, D1 inside U1's outline) - invisible, not merely ambiguous, which defeats the reason I gave. Reclassified as a required P9 move_text fix. Decision revised on file.
- Carried to P9/P10: J3/J4 per-pin silk labels (promised by blocks.md, absent);
- CPL rotation checked per polarized part incl. U3 (DFN-4, reviewer addition);
- JLC order remark to keep break-off tabs off the sensor tongue edge.

Gate attempt history

Timestamp	Gate	Status	Failing/Total
2026-08-27T09:50:50	erc	pass	0/0
2026-08-27T13:18:16	erc	pass	0/0
2026-08-27T15:06:11	place	pass	0/0
2026-08-27T15:08:50	place	pass	0/0
2026-08-27T15:15:05	place	pass	0/0
2026-08-27T15:47:39	place	pass	0/0
2026-08-27T15:57:05	place	pass	0/0
2026-08-27T17:37:18	erc	pass	0/0
2026-08-27T17:37:30	place	pass	0/0
2026-08-27T18:03:33	drc_routed	pass	0/0
2026-08-27T18:09:55	place	pass	0/0
2026-08-27T18:10:01	place	pass	0/0
2026-08-27T18:11:53	verify	fail	5/11
2026-08-27T18:14:54	verify	fail	4/11
2026-08-27T18:38:11	drc_routed	pass	0/0
2026-08-27T18:38:16	verify	fail	1/8
2026-08-27T18:42:14	drc_routed	fail	1/1
2026-08-27T18:43:11	drc_routed	pass	0/0
2026-08-27T18:43:50	drc_routed	pass	0/0
2026-08-27T18:43:51	verify	fail	1/8
2026-08-27T18:49:24	verify	pass	0/8

Timestamp	Gate	Status	Failing/Total
2026-08-27T18:50:05	place	pass	0/0

Issues

Id	Phase	Kinds	Severity	Status
1	P8	insufficient_transition_vias, pour_neckdown	error	fixed
2	P8	thermal_area	error	fixed

Human checkpoints

Id	Phase	Status	When	Note
1	P2	approved	2026-08-27T08:44:13	delegated (unattended): architecture + 2L stackup approved; LDO 0.51 W design point approved; boot-strap gap accepted because the 10k BOOT0 pull-down removes the dependency; outline stays an output; no sim bench
2	P4	approved	2026-08-27T13:21:44	delegated (unattended): schematic approved (erc 0/0, netlist_audit 0, adversarial review 0 errors/3 warnings); W1 Qwiic-ESD waived on ecosystem norm + 2 kV HBM endpoints + indoor bench env; W2 SHT4x VDD slew closed as accepted risk (follower LDO makes I/C unreachable, failure mode is a benign cold-start reset); W3 native DNP on J3/J4 FIXED not waived; P5 proceeds with -outline auto
3	P6	approved	2026-08-27T15:59:22	delegated (unattended): placement approved - gate place 0/0, DRC 0 non-unconnected, size earned 35.79 x 28.34 mm (excess is the SHT40 isolation zone), all four M2 holes kept at zero outline cost, silk 13 -> 0 via the new place.edit silk_clear op plus matching library edits; VBUS pad escape deferred to P7 route-critical with three options costed
4	P8	not reached		
5	P10	not reached		

10 Artifact Index

File	Size	Note
state.json	85,495 B	
kicad/g0-sense.kicad.pcb	250,884 B	
kicad/g0-sense.kicad.sch	1,710 B	
brief/brief.md	1,634 B	
requirements.md	6,653 B	
architecture/blocks.md	11,799 B	
architecture/constraints.json	8,406 B	
architecture/decisions.md	7,408 B	

File	Size	Note
architecture/power_tree.md	4,040 B	
architecture/sheets.md	2,964 B	
architecture/stackup.md	2,679 B	
reports/board_edit_fit.json	1,674 B	
reports/board_edit_slits.json	1,763 B	
reports/drc_p6_final.json	46,582 B	
reports/drc_wol_refill.json	564 B	
reports/erc-waivers.md	6,030 B	
reports/erc.json	1,039 B	
reports/g0-sense_bottom.png	74,299 B	
reports/g0-sense_top.png	276,539 B	
reports/place.json	1,545 B	
reports/place_anneal.json	4,224 B	
reports/place_edit_j1flush.json	245 B	
reports/place_edit_p6_layout.json	3,395 B	
reports/place_edit_p6_silk.json	2,629 B	
reports/place_edit_silkfix.json	757 B	
reports/place_metrics_p6_final.json	6,517 B	
reports/place_seed.json	2,539 B	
reports/review-board.json	3,275 B	
reports/review-board.md	12,824 B	
reports/review-schematic.json	1,621 B	
reports/review-schematic.md	7,856 B	
reports/route_probe_p6.json	286 B	
reports/route_probe_p6_final.json	286 B	
reports/route_probe_p6b.json	286 B	
reports/schematic.pdf	131,338 B	
reports/silk_place.json	18,222 B	
reports/silk_place_p6_final.json	3,347 B	
reports/top.net	60,717 B	
reports/verify-waivers.json	4,600 B	